

BC660K-GL

Hardware Design

NB-IoT Module Series

Version: 1.5

Date: 2025-06-12

Status: Released



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Safety Information

The following safety precautions must be observed during all phases of operation, such as usage, service or repair of any terminal or mobile incorporating the module. Manufacturers of the terminal should notify users and operating personnel of the following safety information by incorporating these guidelines into all manuals of the product. Otherwise, Quectel assumes no liability for customers' failure to comply with these precautions.



Full attention must be paid to driving at all times in order to reduce the risk of an accident. Using a mobile while driving (even with a handsfree kit) causes distraction and can lead to an accident. Please comply with laws and regulations restricting the use of wireless devices while driving.



Switch off the terminal or mobile before boarding an aircraft. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. If there is an Airplane Mode, it should be enabled prior to boarding an aircraft. Please consult the airline staff for more restrictions on the use of wireless devices on an aircraft.



Wireless devices may cause interference on sensitive medical equipment, so please be aware of the restrictions on the use of wireless devices when in hospitals, clinics or other healthcare facilities.



Terminals or mobiles operating over radio signal and cellular network cannot be guaranteed to connect in certain conditions, such as when the mobile bill is unpaid or the (U)SIM card is invalid. When emergency help is needed in such conditions, use emergency call if the device supports it. In order to make or receive a call, the terminal or mobile must be switched on in a service area with adequate cellular signal strength. In an emergency, the device with emergency call function cannot be used as the only contact method considering network connection cannot be guaranteed under all circumstances.



The terminal or mobile contains a transceiver. When it is ON, it receives and transmits radio frequency signals. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.



In locations with explosive or potentially explosive atmospheres, obey all posted signs and turn off wireless devices such as mobile phone or other terminals. Areas with explosive or potentially explosive atmospheres include fueling areas, below decks on boats, fuel or chemical transfer or storage facilities, and areas where the air contains chemicals or particles such as grain, dust or metal powders.

About the Document

Revision History

Version	Date	Author	Description
-	2020-09-30	Clifton HE/ Ellison WANG/ Randy LI	Creation of the document
1.0	2021-01-12	Clifton HE/ Ellison WANG/ Randy LI	First official release
1.1	2021-06-03	Clifton HE/ Ellison WANG/ Randy LI	- Deleted the relevant part of B14. - Chapter 1: Added table 1. - Chapter 2.2: Updated the status of MQTT protocol. - Chapter 5.1: Added this section. - Chapter 7.1 and 7.2: Updated the requirements on storage, manufacturing and soldering.
1.2	2022-09-09	Winks WANG/ Ellison WANG/ Randy LI	- Deleted PDU mode in SMS feature (Table 3). - Added the load current of DC characteristics for VDD_EXT (Table 5). - Added information about ultrasonic cleaning (Chapter 7.2).
1.3	2023-06-02	Weida LIU/ Hugh ZHANG/ Yance YANG	- Added the network protocols CoAP, CoAPS, HTTP and HTTPS (Table 2). - Updated the output power supply (VDD_EXT) from 1.8/3.3 V to 1.8 V/3.1–3.3 V. - Added TVS selection parameter requirements (Table 9). - Added voltage stability requirements (Chapter 3.5.2). - Added the truth table of GRFC interfaces (Table 20). - Updated the recommended reflow soldering

			thermal profile and the relevant parameters. (Figure 32 and Table 34). 7. Updated the recommended reflow soldering thermal profile and the relevant parameters. (Figure 32 and Table 34).
1.4	2024-12-27	Lynne WEI/ Hugh ZHANG/ Kiki JIANG	1. Added comment about adding test point to pin VBAT, VDD_EXT, BOOT, RESET_N, MAIN_TXD/RXD, DBG_TXD/RXD (Table 4). 2. Added comment about power domain to pin RESET_N and PSM_EINT (Table 4). 3. Added relevant notes on pin RESET_N/PSM_EINT. (Chapter 3.2). 4. Added a note of recommending to use AT command before powering off the module (Chapter 3.6.2). 5. Updated the data of parasitic capacitance of TVS array from 50 pF to 15 pF, and added a note of connecting the GND of the USIM card connector to ground (Chapter 3.8). 6. Updated reference circuit for RF antenna interface and added relevant note (Chapter 4.3). 7. Updated the typical and maximum data of power consumption in normal mode (Table 31). 8. Updated the module's coplanarity requirement (Chapter 6.1). 9. Updated the reference document for recommended module stencil thickness (Chapter 7.2). 10. Added a note of avoid using materials that contain mercury (Hg), prohibiting storage or use of unprotected modules in environments containing corrosive gases (Chapter 7.2).
1.5	2025-06-12	Claire LI/ Kun WANG/ Kiki JIANG	1. Added a comment regarding the support for USIM cards under different VBAT conditions and maximum supply current of the USIM_VDD pin (Table 4 & Table 14) 2. Added module's software reset method (Chapter 3.6.3). 3. Updated the note of reference circuit for RF antenna interface (Chapter 4.3). 4. Added a note about the impedance condition for power consumption test (Chapter 5.3). 5. Updated the pre-baking time to 24 h (Chapter

-
- 7.1).
 - 6. Updated the chapter of packaging specification (Chapter 7.3).
-

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1 Introduction

This document defines BC660K-GL and describes its air interface and hardware interfaces which are connected with your applications.

With this document, you can quickly understand module interface specifications, electrical and mechanical details, as well as other related information of the module. The document, coupled with application notes and user guides, makes it easy to design and set up mobile applications with the module.

2 Product Overview

2.1. Frequency Bands and Functions

BC660K-GL is a high-performance NB-IoT module with extremely low power consumption. It is designed to communicate with infrastructures of mobile network operators through NB-IoT radio protocols supported by 3GPP Rel-13 and Rel-14. BC660K-GL supports a broad range of frequency bands as listed below.

Table 1: Frequency Bands of BC660K-GL

Mode	Frequency Bands
LTE HD-FDD	B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/B20/B25/B28/B66/B70/B85

BC660K-GL is an SMD type module with LCC and LGA package, and has an ultra-compact profile of 17.7 mm × 15.8 mm × 2.0 mm, which makes it easily embedded into size-constrained applications and provide reliable connectivity with the applications.

BC660K-GL provides abundant external interfaces (UARTS, ADC, USIM and so on) and protocol stacks (UDP/TCP/LwM2M/MQTT, and so on), which facilitate the module's application.

The module's compact form factor, ultra-low power consumption and extended temperature range make it one of the best choices for a wide range of IoT application. It also provides a complete range of SMS and data transmission services to meet various user demands.

2.2. Key Features

The following table describes the detailed features of BC660K-GL.

Table 2: Key Features

Feature	Details
Power Supply	VBAT: - Supply voltage: 2.2–4.3 V - Typical supply voltage: 3.3 V
Power Saving	Average power consumption (in Deep Sleep mode): 0.8 μ A
Frequency Bands	LTE Cat NB2: B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/B20/B25/B28/B66/B70/B85
Transmitting Power	23 dBm $\pm$ 2.7 dB
eSIM ¹	Reserved with 2.552 $\times$ 2.722 mm package
UART Interfaces	Main UART Interface: - Used for AT command communication and data transmission, where the baud rate is 115200 bps by default. For more details, see Chapter 3.7.1. - Used for firmware upgrade, where the baud rate is 921600 bps by default. Debug UART Interface: - Used for software debugging - Default baud rate: 6 Mbps
USIM Interface	Supports 1.8/3.0 V USIM card
Network Protocols	UDP/TCP/PING/LwM2M/SNTP/MQTT/MQTTS/SSL/TLS/CoAP/CoAPS/HTTP/HTTPS
SMS	- Text Mode - Point-to-point MO and MT
Data Transmission Features	- Single-tone (max.): 25.5 kbps (DL)/16.7 kbps (UL) - Multi-tone (max.): 127 kbps (DL)/158.5 kbps (UL)
AT Commands	3GPP TS 27.005/3GPP TS 27.007 AT commands (3GPP Rel-13) - Quectel enhanced AT commands
Firmware Upgrade	- Main UART interface - DFOTA
Real Time Clock	Supported

¹ eSIM is reserved and not included by default.

Physical Characteristics	- Size: $(17.7 \pm 0.15) \text{ mm} \times (15.8 \pm 0.15) \text{ mm} \times (2.0 \pm 0.2) \text{ mm}$ - Weight: $1.0 \pm 0.2 \text{ g}$
Temperature Range	- Operating temperature range: $-35 \text{ to } +75 \text{ }^{\circ}\text{C}$ ² - Extended temperature range: $-40 \text{ to } +85 \text{ }^{\circ}\text{C}$ ³ - Storage temperature range: $-40 \text{ to } +90 \text{ }^{\circ}\text{C}$
Antenna Interface	50 Ω impedance
RoHS	All hardware components are fully compliant with EU RoHS directive

2.3. Functional Diagram

The following figure shows a block diagram of the module and illustrates the major functional parts.

- RF Transceiver and Subsystem
- Baseband
- Power Management Unit
- Peripheral Interfaces

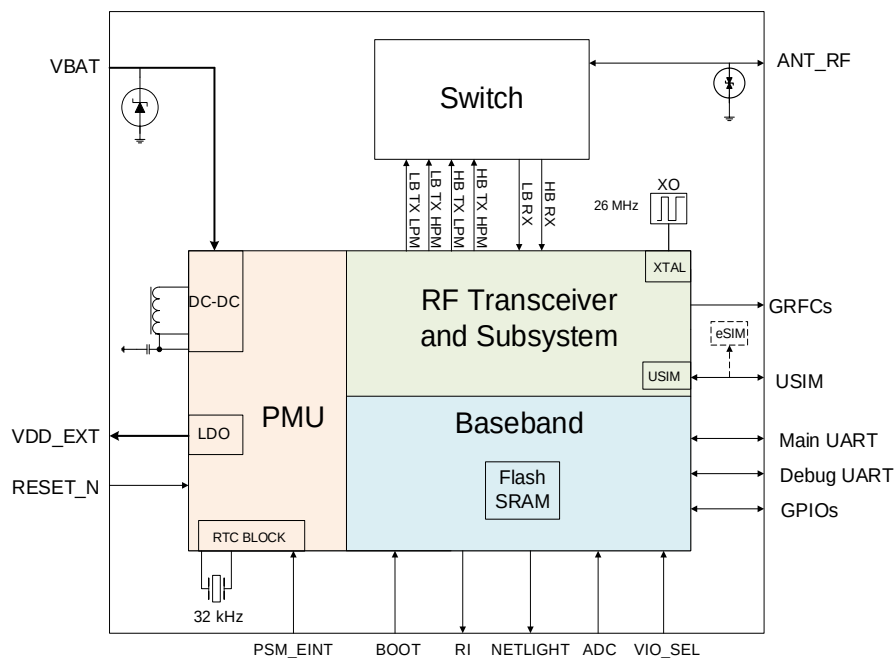


Figure 1: Functional Diagram

² Within this range, the module's indicators comply with 3GPP specification requirements.

³ Within this range, the module retains the ability to establish and maintain functions such as SMS and data transmission, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out} , may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

2.4. TE-B Kit

Quectel supplies an evaluation board (BC660K-GL TE-B) with accessories to develop and test the module. For more details, see ***document [1]***.

3 Application Functions and Interfaces

The module is equipped with 58 pins, including 44 LCC pins and 14 LGA pins. The subsequent chapters provide detailed descriptions of the following functions/pins/interfaces:

- Power Supply
- PSM_EINT
- RESET_N
- BOOT
- UART Interfaces
- USIM Interface
- ADC Interface
- RI
- GPIO interfaces
- GRFC interfaces
- NETLIGHT

3.1. Pin Assignment

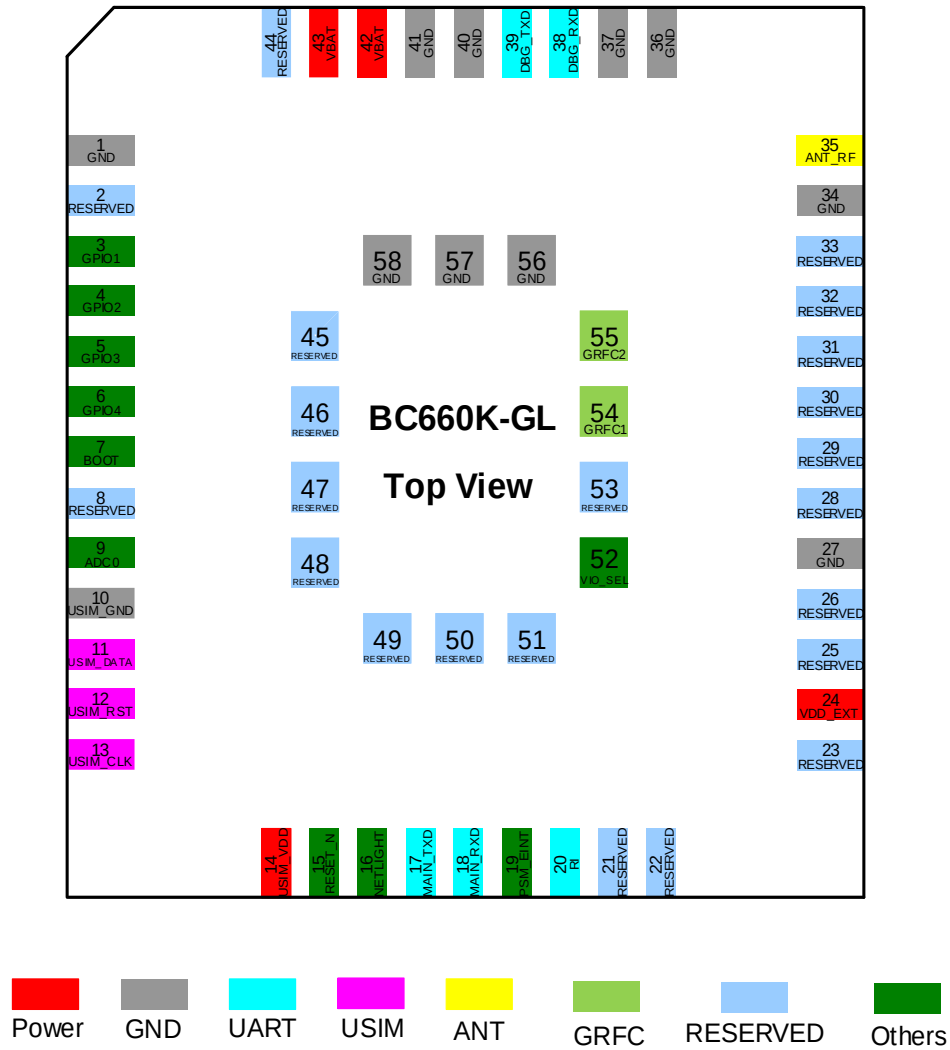


Figure 2: Pin Assignment

NOTE

Keep all RESERVED and unused pins unconnected.

3.2. Pin Description

Table 3: I/O Parameter Definition

Type	Description
AI	Analog Input
AIO	Analog Input/Output
DI	Digital Input
DO	Digital Output
DIO	Digital Input/Output
PI	Power Input
PO	Power Output

DC characteristics include power domain and rated current.

Table 4: Pin Description

Power Supply					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
VBAT	42, 43	PI	Power supply for the module	Vmax = 4.3 V Vmin = 2.2 V Vnom = 3.3 V	Test points are recommended to be reserved.
VDD_EXT	24	PO	1.8 V/3.1–3.3 V output power supply (Controlled by VIO_SEL, 1.8 V by default)	Vnom = 1.8 V/3.1–3.3 V Iomax = 40 mA	No voltage output in Deep/Light Sleep mode. It is intended to supply power for the module's pull-up circuits, and is not recommended to supply power for external circuits. A test point is recommended to be

reserved.

GND 1, 27, 34, 36, 37, 40, 41, 56–58

BOOT

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
BOOT	7	DI	Make the module enter download mode	$V_{ILmax} = 0.2 \times VDD_EXT$ $V_{IHmin} = 0.7 \times VDD_EXT$	Active low. A test point is recommended to be reserved.

Reset

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
RESET_N	15	DI	Reset the module	$V_{ILmax} = 0.38\text{ V}$ $V_{IHmin} = 1.47\text{ V}$ $V_{IHmax} = 2.1\text{ V}$	1.9–2.1 V power domain. Active low. A test point is recommended to be reserved if unused.

PSM_EINT Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
PSM_EINT	19	DI	External interrupt pin dedicated to waking up the module from Deep/Light Sleep mode.	$V_{ILmax} = 0.38\text{ V}$ $V_{IHmin} = 1.47\text{ V}$ $V_{IHmax} = 2.1\text{ V}$	1.9–2.1 V power domain. Active on falling edge.

Network Status Indication

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
NETLIGHT	16	DO	Indicate the module's network activity status	$V_{OLmax} = 0.15 \times VDD_EXT$ $V_{OHmin} = 0.8 \times VDD_EXT$	

ADC Interface

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
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ADC0	9	AI	General-purpose ADC interface	Input Voltage range: 0–1.2 V	
Main UART Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
MAIN_RXD	18	DI	Main UART receive	$V_{ILmax} = 0.2 \times VDD_EXT$ $V_{IHmin} = 0.7 \times VDD_EXT$	Test points are recommended to be reserved.
MAIN_TXD	17	DO	Main UART transmit	$V_{OLmax} = 0.15 \times VDD_EXT$ $V_{OHmin} = 0.8 \times VDD_EXT$	
Debug UART Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
DBG_RXD	38	DI	Debug UART receive	$V_{ILmax} = 0.2 \times VDD_EXT$ $V_{IHmin} = 0.7 \times VDD_EXT$	Test points must be reserved.
DBG_TXD	39	DO	Debug UART transmit	$V_{OLmax} = 0.15 \times VDD_EXT$ $V_{OHmin} = 0.8 \times VDD_EXT$	
RI					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
RI	20	DO	Ring indication	$V_{OLmax} = 0.15 \times VDD_EXT$ $V_{OHmin} = 0.8 \times VDD_EXT$	
USIM Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
USIM_VDD	14	PO	USIM card power supply	$V_{nom} = 1.8/3.0\text{ V}$	When $3.0\text{ V} \leq VBAT \leq 4.3\text{ V}$, 1.8/3.0 V USIM card is supported; When $2.2\text{ V} \leq VBAT < 3.0\text{ V}$, only 1.8 V USIM card is supported. Maximum supply current: about 80 mA.
USIM_CLK	13	DO	USIM card clock	$V_{OLmax} = 0.15 \times USIM_VDD$ $V_{OHmin} = 0.8 \times USIM_VDD$	
USIM_RST	12	DO	USIM card reset	$V_{OLmax} = 0.15 \times USIM_VDD$ $V_{OHmin} = 0.8 \times USIM_VDD$	

USIM_DATA	11	DIO	USIM card data	V _{IL} max = 0.2 × USIM_VDD V _{IH} min = 0.7 × USIM_VDD V _{OL} max = 0.15 × USIM_VDD V _{OH} min = 0.8 × USIM_VDD	
USIM_GND	10		Dedicated ground for USIM card		
Antenna Interface					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
ANT_RF	35	AIO	RF antenna interface		50 Ω characteristic impedance.
GPIO Interfaces					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
GPIO1	3	DIO	General-purpose input/output	V _{IL} max = 0.2 × VDD_EXT V _{IH} min = 0.7 × VDD_EXT V _{OL} max = 0.15 × VDD_EXT V _{OH} min = 0.8 × VDD_EXT	If unused, keep these pins open.
GPIO2	4	DIO	General-purpose input/output		
GPIO3	5	DIO	General-purpose input/output		
GPIO4	6	DIO	General-purpose input/output		
GRFC Interfaces					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
GRFC1	54	DO	Generic RF controller	V _{OL} max = 0.27 V V _{OH} min = 1.44 V	1.8 V power domain.
GRFC2	55	DO	Generic RF controller	V _{OL} max = 0.27 V V _{OH} min = 1.44 V	If unused, keep these pins open.
Others					
Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
VIO_SEL	52	DI	I/O voltage		Control VDD_EXT

selection

voltage selection ⁴

Reserved Pins

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
RESERVED	2, 8, 21–23, 25, 26, 28–33, 44–51, 53				Keep these pins open.

NOTE

1. The RESET_N/PSM_EINT pin has an internal design to prevent current backflow, so the voltage of these pins measured with a multimeter is not the real voltage inside the chip.
2. When VIO_SEL is grounded and VBAT ≥ 3.3 V, VDD_EXT does not output the exact typical voltage of 3.3 V, the actual output voltage is between 3.1–3.3 V.
3. Keep all reserved and unused pins unconnected.

3.3. Operating Modes

The following table briefly describes the three working modes of the module.

Table 5: Application Processor (AP) Operating Modes

Mode	Description
Normal	In normal mode, the AP handles tasks such as AT command communication.
Idle	When all tasks are suspended, the AP enters idle mode.

Table 6: Modem Operating Modes

Mode	Description
Connected	The network is connected and the module supports data transmission. In such a case, the modem can switch to DRX/eDRX mode.
DRX/eDRX	The modem is in idle mode, and downlink data can be received during PTW only. In such a case, the modem can switch to PSM or connected mode.

⁴ When VIO_SEL is grounded and VBAT < 3.3 V, VDD_EXT = VBAT; When VIO_SEL is grounded and VBAT ≥ 3.3 V, VDD_EXT = 3.1–3.3 V; When VIO_SEL is floating, VDD_EXT = 1.8 V.

PSM	In power saving mode, the modem is disconnected from the network and cannot receive any downlink data. In such a case, the modem can switch to eDRX/DRX mode, then quickly switch to connected mode.
-----	--

Table 7: Module Operating Modes

Mode	Description
Active	When the AP is in normal mode and the modem is in connected mode, the module is active and supports all services and functions. The current consumption in active mode is higher than that in sleep modes.
Light Sleep	Generally, when the AP is in idle mode and the modem is not in PSM mode, the module enters Light Sleep mode. In Light Sleep mode, the current consumption of the module is reduced greatly.
Deep Sleep	When the AP is in idle mode and the modem is in PSM, the module enters Deep Sleep mode in which the CPU is powered off and only the 32 kHz RTC clock keeps working. In Deep Sleep mode, the current consumption is minimized (average value: 0.8 μ A).

3.4. Power Saving Modes

3.4.1. Light Sleep Mode

In Light Sleep mode, the UART interfaces do not work, and the module can be woken up on the falling edge of PSM_EINT or by sending the AT command to it via the main UART interface (this AT command will be lost).

3.4.2. Deep Sleep Mode

The module consumes extremely low current in Deep Sleep mode (average value: 0.8 μ A). The main purpose of Deep Sleep is to reduce the power consumption of the module and prolong the power supply duration of the battery. In this mode, the UART interfaces do not work.

The following figure shows the power consumption of the module in different modes.

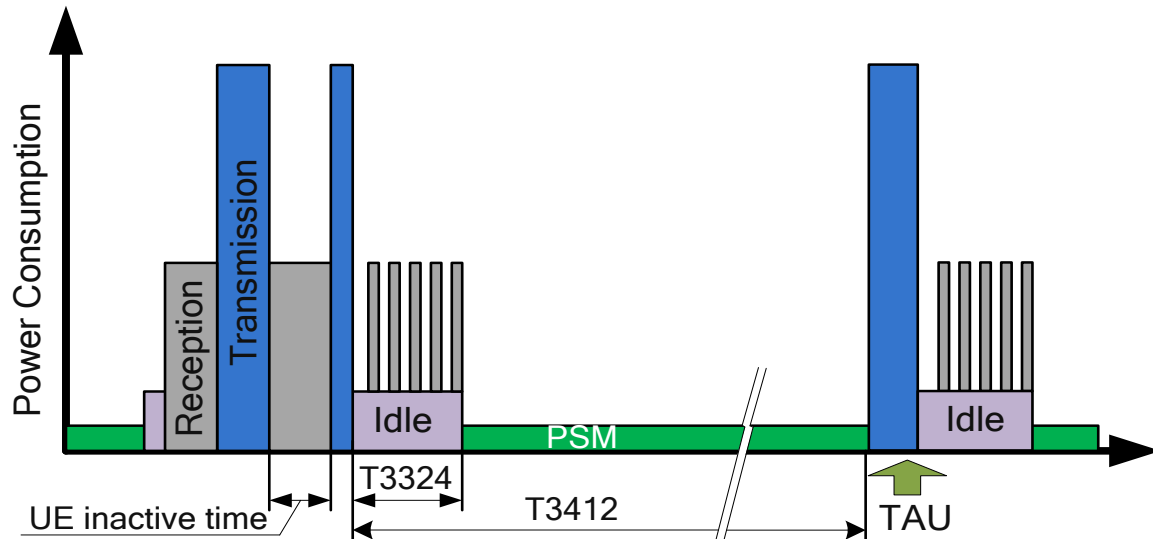


Figure 3: Module Power Consumption in Different Modem Modes

When the modem remains in PSM and the AP is in idle mode, the module will enter Deep Sleep mode. The procedure of the modem entering PSM is as follows:

The modem requests to enable PSM with an **ATTACH REQUEST** or **TAU REQUEST** message during ATTACH or TAU (Tracking Area Update) procedure. Then the network accepts the request and provides an active time value (T3324) to the modem and the mobile reachable timer starts. When the T3324 timer expires, the modem enters PSM. Please note that the module cannot request entering PSM when establishing an emergency attachment or initializing the PDN (Public Data Network) connection.

When the module is in Deep Sleep mode, it can be woken up in the following cases:

- After the T3412 timer expires, the module will exit from Deep Sleep automatically.
- After the MCU sends an AT command to the module (this AT command will be lost), and pulls down the MAIN_RXD, the module will be woken up from Deep Sleep in MAIN_RXD's falling edge.
- Pulling down PSM_EINT (falling edge) will wake up the module from Deep Sleep.

The timing of waking up the module from Deep Sleep is illustrated below.

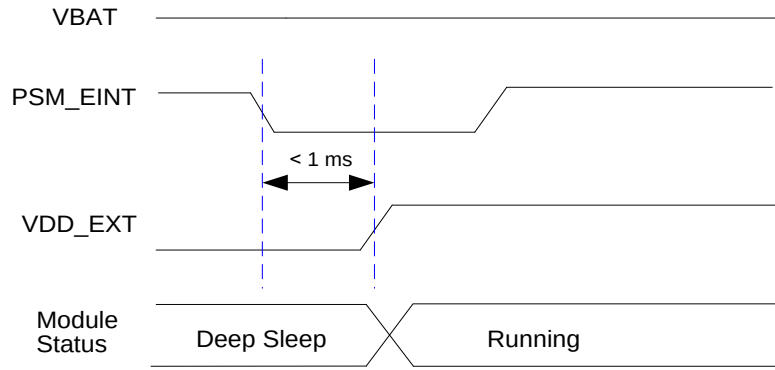


Figure 4: Timing of Waking Up Module from Deep Sleep

3.5. Power Supply

3.5.1. Power Supply Pins

The module provides two VBAT pins for connection with an external power supply. The table below describes the module's VBAT and ground pins.

Table 8: Power Supply Pins

Pin Name	Pin No.	Description	Min.	Typ.	Max.	Unit
VBAT	42, 43	Power supply for the module	2.2	3.3	4.3	V
GND	1, 27, 34, 36, 37, 40, 41, 56, 57, 58					

3.5.2. Voltage Stability Requirements

The power supply range of the module is from 2.2 V to 4.3 V. Make sure that the input voltage will never drop below 2.2 V. The following figure shows the voltage drop during heavy load.

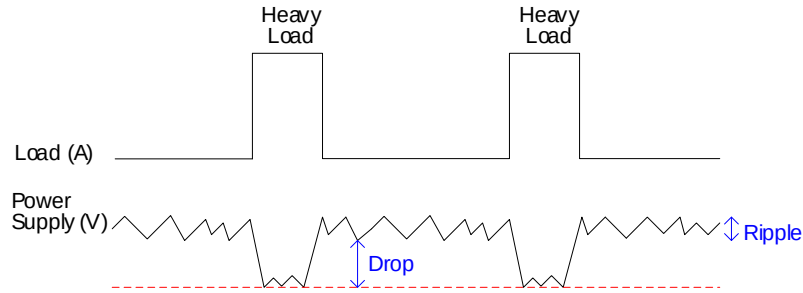


Figure 5: Power Supply Limits during Heavy Load

3.5.3. Power Supply Reference Design

Power design for a module is critical to its performance. It is recommended to use a low quiescent current LDO with an output capacity of 0.5 A to regulate the power supply for BC660K-GL. Lithium-thionyl chloride (Li-SOCl₂) batteries and Lithium manganese oxide (LiMn₂O₄) batteries can be used as the power supply. The supply voltage of the module ranges from 2.2 V to 4.3 V. When the module is working, ensure its input voltage never drops below 2.2 V; otherwise, the module cannot work normally.

For better power performance, it is recommended to place a 100 μ F capacitor with low ESR (ESR = 0.7 Ω) and three ceramic capacitors (100 nF, 100 pF and 22 pF) near the VBAT pins. Also, it is recommended to add a TVS on the VBAT trace (near VBAT pins) to improve surge voltage withstanding capability. In principle, the longer the VBAT trace is, the wider it should be. A reference circuit for power supply is illustrated in the following figure.

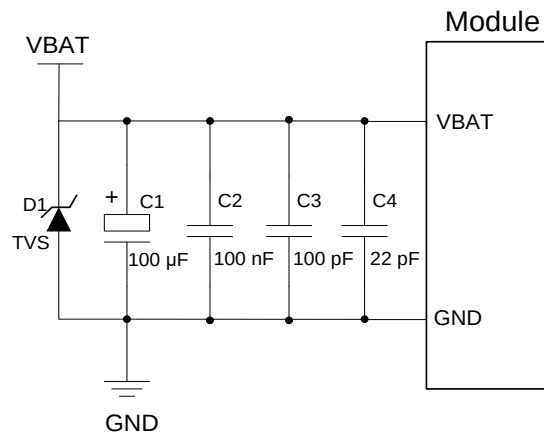


Figure 6: Reference Design for Power Supply

Table 9: TVS Selection Parameter Requirements

Symbol	Parameter	Min.	Typ.	Max.	Unit	Test Condition
V_{RWM}	Reverse operating voltage	-	-	4.85	V	-
V_{BR}	Breakdown voltage	4.9	5.3	6	V	$I_{BR} = 1 \text{ mA}$
V_C	Clamping voltage	-	5.5	-	V	$I_{PP} = 16 \text{ A}$ ($t_P = 100 \text{ ns}$)
V_C	Clamping voltage	-	5.5	-	V	$I_{PP} = 10 \text{ A}$ ($8 \times 20 \mu\text{s}$ pulse)
V_C	Clamping voltage	-	9.5	-	V	$I_{PP} = 180 \text{ A}$ ($8 \times 20 \mu\text{s}$ pulse)
V_C	Clamping voltage	-	7.5	-	V	$V_{ESD} = 8 \text{ kV}$
I_R	Reverse leakage current	-	-	1.0	μA	$V_{RWM} = 4.85 \text{ V}$

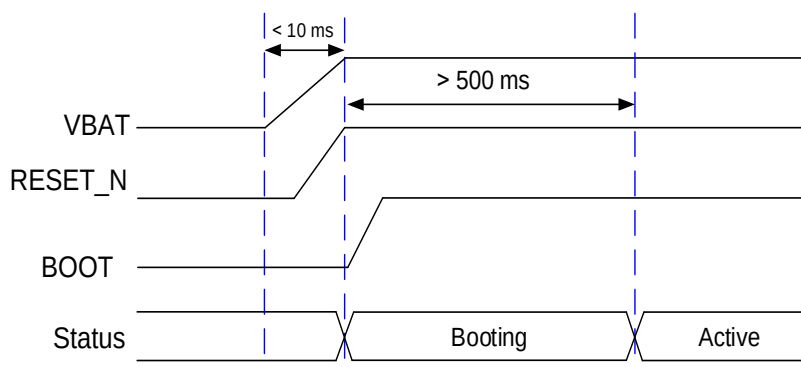
3.5.4. Power Supply Voltage Monitoring

You can use **AT+CBC** to monitor and query the current VBAT voltage. The unit of the voltage value is mV (millivolt). For detailed information about the command, see **document [2]**.

3.6. Turn-on/Turn-off Scenario

3.6.1. Turn On

After the module VBAT is powered up, keep the RESET_N and BOOT high (both pins are at high level by default), and the module will turn on automatically. The turn-on timing is illustrated in the following figure.


Figure 7: Turn-on Timing

NOTE

1. After cutting off the VBAT, the module can be turned on again only after its VBAT voltage drops below 0.7 V. The actual discharging time of VBAT needs to be determined based on circuit tests and enough time margin should be left to avoid turning on abnormally.
2. The power-up time of VBAT must be within 10 ms, and after VBAT is powered up, RESET_N and BOOT automatically rise to high level due to internal pull-ups.
3. It is recommended that the MCU retains a RESET_N controlling pin so that the RESET_N can reset the module to exit the abnormal state when the module is operating abnormally due to the abnormal power-up sequence or dropping supply voltage of VBAT.

3.6.2. Turn Off

The module can be turned off through cutting off its VBAT power supply.

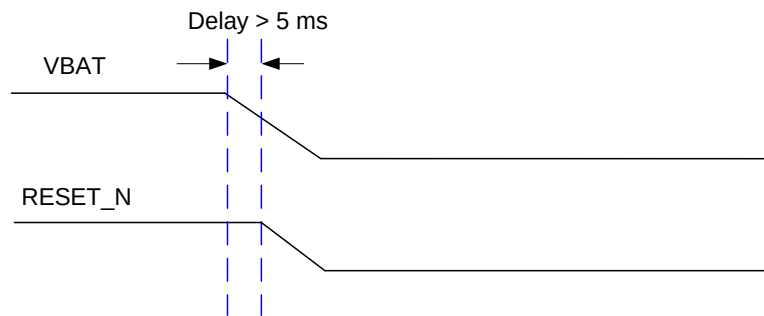


Figure 8: Turn-off Timing

NOTE

It is recommended to execute **AT+CFUN=0** and return to **OK** before powering down the module. For detailed information about the command, see *document [2]*.

3.6.3. RESET_N

The module can be reset through the following methods:

- Hardware reset: driving RESET_N low for at least 50 ms will reset the module.
- Software reset: sending **AT+QRST**. For more details about the AT command, see *document [2]*.

Table 10: Pin Description of RESET_N

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
RESET_N	15	DI	Reset the module.	$V_{ILmax} = 0.38\text{ V}$ $V_{IHmin} = 1.47\text{ V}$ $V_{IHmax} = 2.1\text{ V}$	1.9–2.1 V power domain. Active low. A test point is recommended to be reserved if unused.

The reference designs for resetting the module are shown below. An open drain/collector driving circuit or a button can be used to control the RESET_N pin.

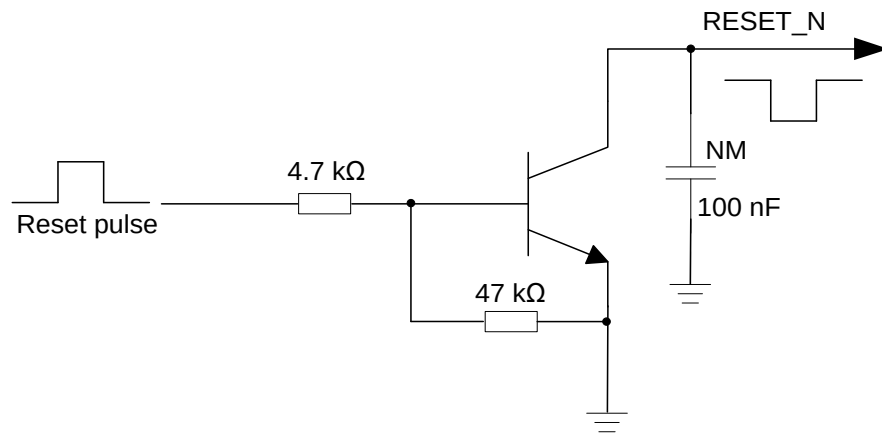


Figure 9: Reference Design for RESET_N Controlled with an OC/OD Driving Circuit

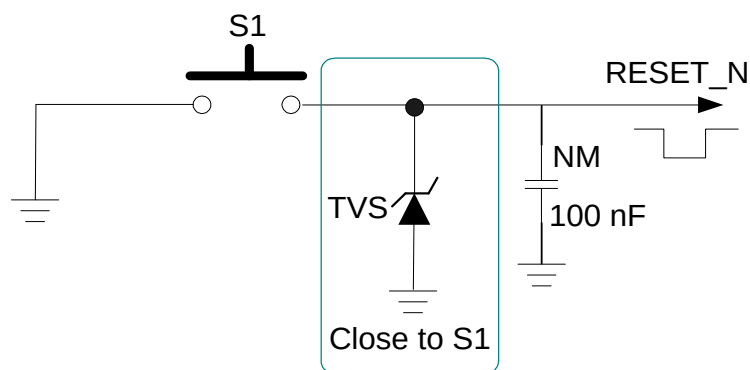


Figure 10: Reference Design for RESET_N Controlled with a Button

NOTE

It is recommended to reserve a 100 nF capacitor position, and the capacitor is not mounted by default.

3.6.4. Download Mode

Table 11: Pin Description of BOOT

Pin Name	Pin No.	I/O	Description	DC Characteristics	Comment
BOOT	7	DI	Make the module enter download mode	$V_{ILmax} = 0.2 \times VDD_EXT$ $V_{IHmin} = 0.7 \times VDD_EXT$	Active low. A test point is recommended to be reserved.

In the process of resetting or powering on the module, drive and keep the BOOT pin low and the module will enter the download mode.

In the download mode, the firmware can be downloaded through the main UART interface. After the download is completed, the module needs to be reset to exit from the download mode.

A reference design is shown below.

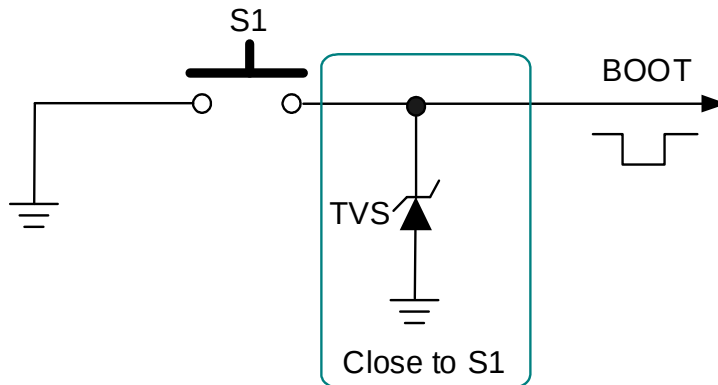


Figure 11: Reference Design for BOOT Controlled with a Button

NOTE

If the BOOT is connected to a filter capacitor in parallel, the capacitance of the capacitor cannot be higher than 33 pF.

3.7. UART Interfaces

The module provides two UART interfaces: the main UART interface and the debug UART interface.

Table 12: Pin Description of UART Interfaces

Interface	Pin Name	Pin No.	I/O	Description	Comment
Main UART Interface	MAIN_TXD	17	DO	Main UART transmit	Test points are recommended to be reserved.
	MAIN_RXD	18	DI	Main UART receive	
Debug UART Interface	DBG_RXD	38	DI	Debug UART receive	Test points must be reserved.
	DBG_TXD	39	DO	Debug UART transmit	
Ring Indication	RI	20	DO	Ring indication	

3.7.1. Main UART Interface

The main UART interface supports AT command communication, data transmission and firmware upgrade.

- Default baud rate: 115200 bps
- Supported baud rates: 2400 bps, 4800 bps, 9600 bps, 19200 bps, 38400 bps, 57600 bps, 115200 bps, 230400 bps, 460800 bps

When the interface is used for firmware upgrade, the baud rate is 921600 bps by default.

When the module is in Deep Sleep/Light Sleep mode, the MCU can wake up it by sending AT commands through the main UART interface and meanwhile the MCU pulls down the MAIN_RXD. The module will be woken up from Deep Sleep in MAIN_RXD's falling edge. It is recommended to keep sending the AT command until **OK** is returned before sending AT commands for other services.

The figure below shows the connection between the module and the MCU.

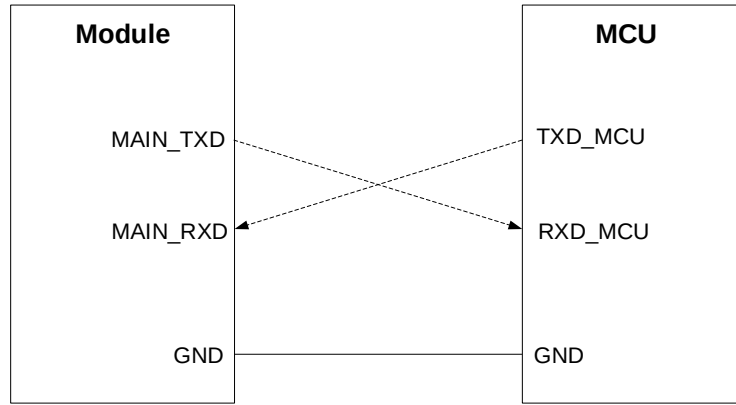


Figure 12: Reference Design for Main UART Interface Connection

3.7.2. Debug UART Interface

Through debug tools, the debug UART interface can be used to output logs for software debugging. Its baud rate is 6 Mbps by default. The following is a reference design of debug UART interface. This UART interface only supports outputting log, and cannot be used for data transmission.

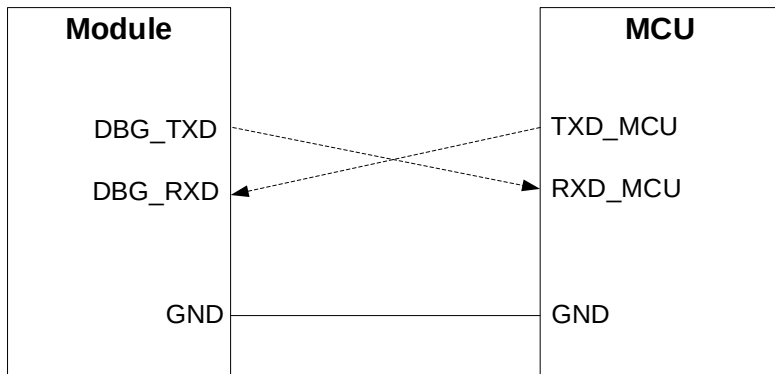


Figure 13: Reference Design for Debug UART Interface Connection

3.7.3. UART Interface Application

Table 13: Pin Description of VIO_SEL

Pin Name	Pin No.	I/O	Description	Comment
VIO_SEL	52	DI	I/O Voltage selection	Control VDD_EXT voltage selection

The UART interface voltage domain of this module is optional. You can select the appropriate voltage domain through VIO_SEL according to actual situation. When VIO_SEL is floating, the VDD_EXT voltage domain is 1.8 V; when VIO_SEL is grounded, the VDD_EXT voltage domain is 3.1–3.3 V or equals to the VBAT voltage ⁵. If the voltage domain of your application system is 1.8 V, VIO_SEL should be floating; If the voltage domain of your application system is 3.1–3.3 V, VIO_SEL must be grounded.

The following figure shows the reference design of UART interfaces:

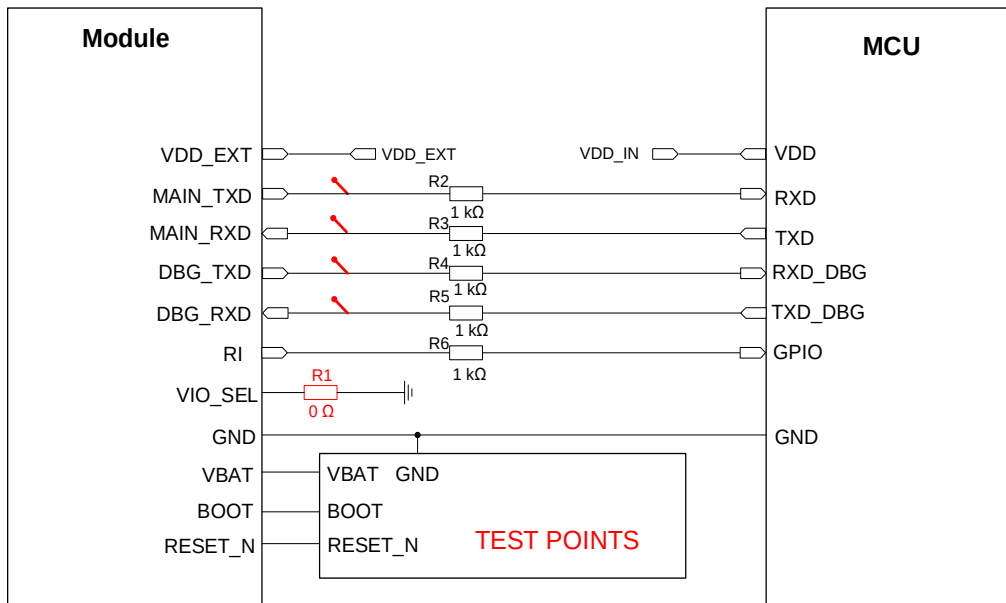


Figure 14: Reference Design for UART Interface Connection

The following circuit shows a reference design for the communication between the module and a PC with a standard RS-232 interface. Make sure to select appropriate voltage domain through VIO_SEL according to the actual situation.

⁵ When VIO_SEL is grounded and VBAT < 3.3 V, VDD_EXT = VBAT; When VIO_SEL is grounded and VBAT ≥ 3.3 V, VDD_EXT = 3.1–3.3 V; When VIO_SEL is floating, VDD_EXT = 1.8 V.

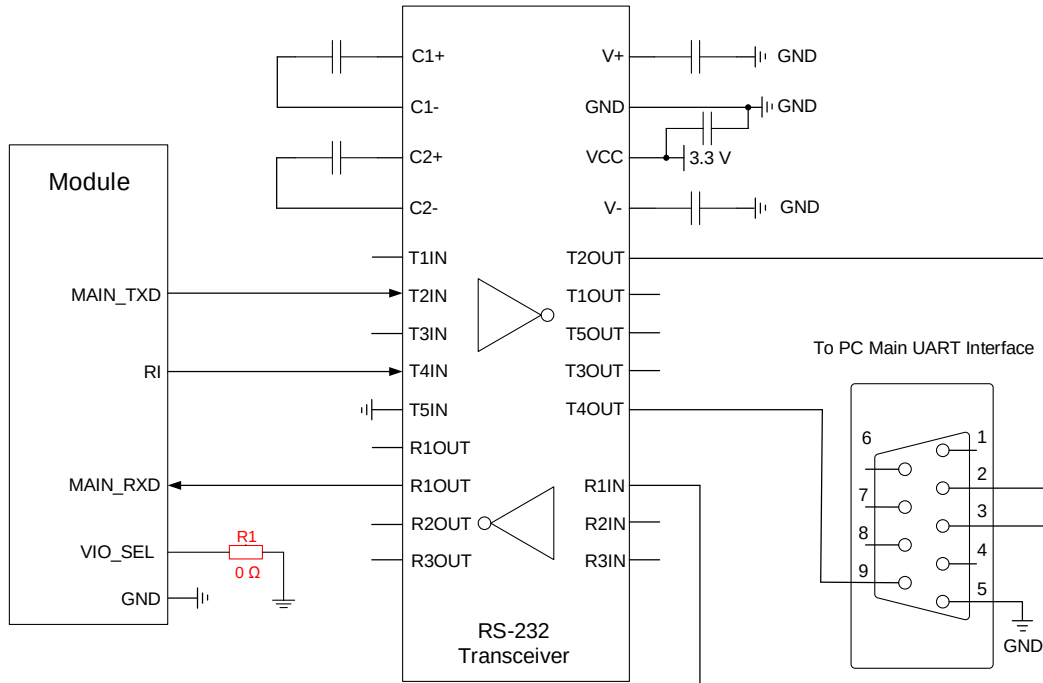


Figure 15: Reference Design for Module-PC Communication via RS-232 Interface

Please visit vendors' websites to select a suitable RS-232 transceiver, such as <http://www.exar.com> and <http://www.maximintegrated.com>.

NOTE

1. If the voltage domain of your application system is 1.8 V, keep the R1 in red not mounted; if it is 3.1-3.3 V, keep the R1 in red mounted.
2. “” represents the test points of UART interfaces. It is recommended to reserve the test points of VBAT, BOOT and RESET_N for convenient firmware upgrade and software debugging when necessary.
3. MAIN_RXD cannot be pulled up to VDD_EXT directly. To pull up MAIN_RXD to VDD_EXT, you need to connect a Schottky diode in series first, and then add a pull-up resistor of 4.7–20 kΩ. For more details, see **document [3]**.

If the voltage domain of your application system is neither 1.8 V nor 3.1–3.3 V, it is recommended to use a level-shifting circuit. For the design of the circuit shown in the dotted line, refer to that shown in the solid lines, and pay attention to the connection direction. In this case, the design of RI–GPIO circuit can refer to that of the MAIN_TXD–RXD circuit.

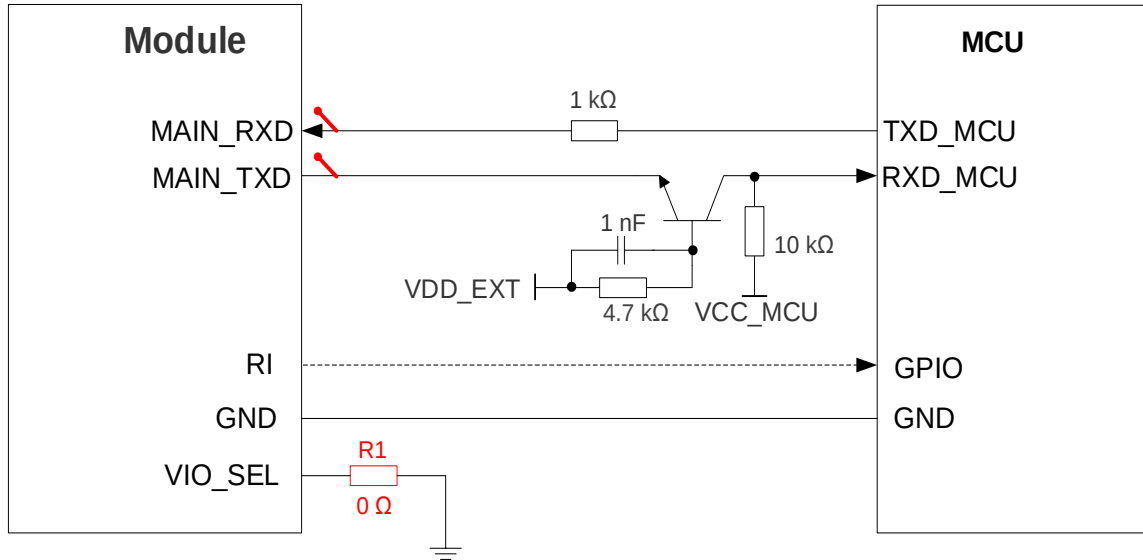


Figure 16: Reference Design for Level-shifting Circuit

NOTE

1. Due to the anti-backflow design of the MAIN_RXD pin, it can be directly connected to the TXD of the peripheral in 1.8–3.3 V voltage domain. If wake-up function of MAIN_RXD in Deep Sleep/Light Sleep mode is enabled, it is recommended that MAIN_RXD not use any level-shifting circuit so as to avoid abnormal wake-up.
2. If you apply the level-shifting circuit, don't mount the R1 marked in red.
3. MAIN_RXD cannot be pulled up to VDD_EXT directly. To pull up MAIN_RXD to VDD_EXT, you need to connect a Schottky diode in series first, and then add a pull-up resistor of 4.7–20 kΩ. For more details, see **document [3]**.
4. Transistor solution is not suitable for applications with baud rates exceeding 460 kbps.

3.8. USIM Interface

The USIM card is powered by USIM_VDD. Both 1.8 V and 3.0 V USIM cards are supported.

Table 14: Pin Description of USIM Interface

Pin Name	Pin No.	I/O	Description	Comment
USIM_VDD	14	PO	USIM card power supply	When $3.0\text{ V} \leq \text{VBAT} \leq 4.3\text{ V}$, 1.8/3.0 V USIM card is supported; When $2.2\text{ V} \leq \text{VBAT} < 3.0\text{ V}$, only 1.8 V USIM card is

supported.

Maximum supply current: about 80 mA.

USIM_CLK	13	DO	USIM card clock
USIM_RST	12	DO	USIM card reset
USIM_DATA	11	DIO	USIM card data
USIM_GND	10		Dedicated ground for USIM card

A reference design for the USIM interface with a 6-pin USIM card connector is below.

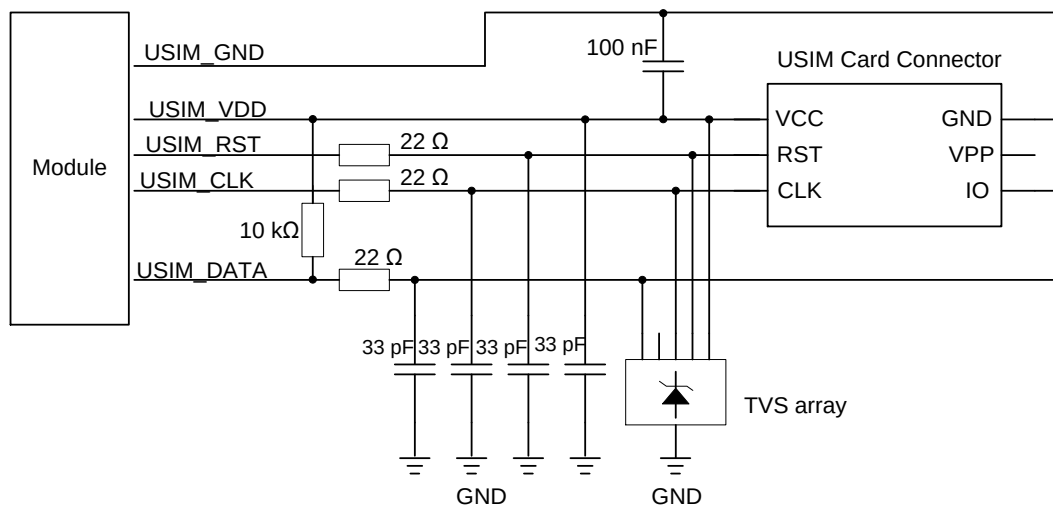


Figure 17: Reference Design for USIM Interface with a 6-pin USIM Card Connector

NOTE

1. If your PCB board has a complete ground plane on the GND between the module and the USIM card connector, you can directly connect the GND of the USIM card connector to the nearest GND plane. Otherwise, it is recommended to connect the GND of the USIM card connector to the USIM_GND, and the shorter the USIM_GND trace is, the better.
2. The pull-up resistor of 10 kΩ on the USIM_DATA can improve anti-jamming capability and should be placed close to the USIM card connector.

To enhance the reliability and availability of the USIM card in applications, follow the criteria below in USIM circuit design:

- Place the USIM card connector as close to the module as possible. Keep the trace length as short as possible, at most 200 mm.
- Keep USIM card signal lines away from RF and power supply traces.

- Make the trace between the ground of the module and that of the USIM card connector short and wide and ensure the trace width not less than 0.5 mm avoid any decrease in electric potential. The decoupling capacitor between USIM_VDD and GND should be not more than 1 μ F and be placed close to the USIM card connector.
- To avoid cross-talk between USIM_DATA and USIM_CLK, keep them away from each other and shield them separately with surrounding ground.
- For better ESD protection, it is recommended to add a TVS array whose parasitic capacitance should be not more than 15 pF. Place the ESD protection component as close to the USIM card connector as possible, and ensure the USIM card signal lines from the USIM card connector go through the ESD protection component before reaching the module. The 22 Ω resistors should be connected in series between the module and the USIM card connector to suppress EMI and enhance ESD protection. Note that the module's USIM peripheral devices should be placed close to the USIM card connector.
- Add 33 pF capacitors in parallel among USIM_DATA, USIM_CLK and USIM_RST signal traces to filter out RF interference.

3.9. ADC Interface

The module provides a 12-bit ADC input channel to read the voltage value.

Table 15: Pin Description of ADC Interface

Pin Name	Pin No.	I/O	Description	Comment
ADC0	9	AI	General-purpose ADC interface	Input voltage range: 0–1.2 V.

NOTE

A 320 k Ω pull-down resistor is integrated inside the ADC0 pin. This resistor needs to be considered when you calculate the voltage division relationship.

3.10. RI

When there is a message received or a URC output, the module will notify the peripheral through the RI interface.

Table 16: Pin Description of RI Interface

Pin Name	Pin No.	I/O	Description	Comment
RI	20	DO	Ring indication	VDD_EXT power domain

Table 17: RI Signal Status

Module Status	RI Signal Level
Idle	High
URC/SMS Message Arrives	Low for at least 120 ms before starting data output.

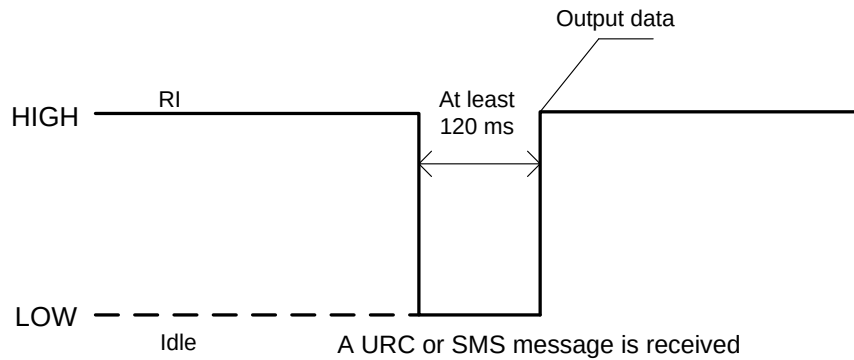


Figure 18: RI Behaviour When a URC/SMS Message Is Received

3.11. GPIO Interfaces

The module provides four general-purpose input and output (GPIO) interfaces. **AT+QCFG="gpio"** command can be used to configure the status of the GPIO pins. For more details about the AT command, see [document \[2\]](#).

Table 18: Pin Description of GPIO Interfaces

Pin Name	Pin No.	I/O	Description	Comment
GPIO1	3	DIO	General-purpose	If unused, keep these pins

GPIO2	4	DIO	input/output	open.
GPIO3	5	DIO		
GPIO4	6	DIO		

3.12. GRFC Interfaces

The module provides two generic RF control interfaces for the control of external antenna tuners.

Table 19: Pin Description of GRFC Interfaces

Pin Name	Pin No.	I/O	Description	Comment
GRFC1	54	DO	Generic RF controller	1.8 V power domain.
GRFC2	55	DO	Generic RF controller	If unused, keep these pins open.

Table 20: Truth Table of GRFC Interfaces

GRFC1 Level	GRFC2 Level	Frequency Range (MHz)	Band
Low	Low	880–2180	B1, B2, B3, B4, B8, B25, B66, B70
Low	High	791–894	B5, B18, B19, B20
High	Low	698–803	B12, B13, B17, B28, B85
High	High	-	Disabled

3.13. NETLIGHT

NETLIGHT can be used to indicate the network status of the module.

Table 21: Pin Description of NETLIGHT

Pin Name	Pin No.	I/O	Description	Comment
NETLIGHT	16	DO	Indicate the module's network activity status	$V_{OLmax} = 0.15 \times VDD_EXT$ $V_{OHmin} = 0.8 \times VDD_EXT$

A reference design for NETLIGHT is shown below.

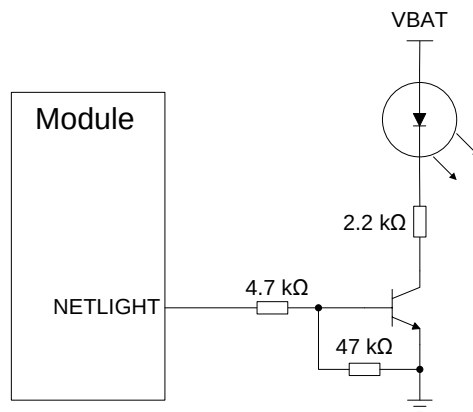


Figure 19: Reference Design for NETLIGHT

4 RF Specifications

Appropriate antenna type and design should be used with matched antenna parameters according to specific application. It is required to perform a comprehensive functional test for the RF design before mass production of terminal products. The entire content of this chapter is provided for illustration only. Analysis, evaluation and determination are still necessary when designing target products.

The pin 35 is the RF antenna pad. The antenna interface has an impedance of 50 Ω .

4.1. Antenna Interface

Table 22: Pin Description of NB-IoT Antenna Interface

Pin Name	Pin No.	I/O	Description
ANT_RF	35	AIO	RF antenna interface
GND	34, 36, 37		Ground

4.2. Operating Frequency

Table 23: Module Operating Frequency

Frequency Band	Receiving Frequency	Transmitting Frequency
B1	2110–2170 MHz	1920–1980 MHz
B2	1930–1990 MHz	1850–1910 MHz
B3	1805–1880 MHz	1710–1785 MHz
B4	2110–2155 MHz	1710–1755 MHz
B5	869–894 MHz	824–849 MHz

B8	925–960 MHz	880–915 MHz
B12	729–746 MHz	699–716 MHz
B13	746–756 MHz	777–787 MHz
B17	734–746 MHz	704–716 MHz
B18	860–875 MHz	815–830 MHz
B19	875–890 MHz	830–845 MHz
B20	791–821 MHz	832–862 MHz
B25	1930–1995 MHz	1850–1915 MHz
B28	758–803 MHz	703–748 MHz
B66	2110–2180 MHz	1710–1780 MHz
B70	1995–2020 MHz	1695–1710 MHz
B85	728–746 MHz	698–716 MHz

4.3. Reference Design

BC660K-GL provides an RF antenna pin for external NB-IoT antenna connection.

- The RF trace on the host PCB should be a coplanar waveguide or microstrip whose characteristic impedance is 50 Ω .
- The module comes with ground pads which are next to the antenna pad to give a better grounding.
- To achieve better RF performance, it is recommended to reserve a dual L-type circuit and place the dual L-type components (R1/C1/C2/C3) as close to the antenna as possible. By default, the capacitors (C1/C2) are not mounted and a 0 Ω resistor is mounted on R1.

A reference design of the RF interface is shown below.

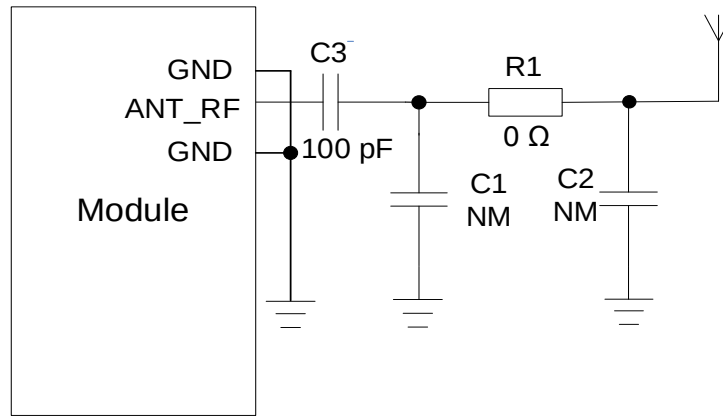


Figure 20: Reference Design for the Antenna Interface

NOTE

Notes on C3:

- 1) If there is DC power at the antenna ports, place capacitor on C3 to prevent short circuit to ground. The capacitance value is recommended to be 100 pF, which can be adjusted according to the debugging results.
- 2) If there is no DC power in the peripheral design:
 - a) Do not reserve C3.
 - b) If C3 has already been reserved, it should be mounted with a component, and it is recommended to use a 0 Ω resistor. You can also match the component according to the debugging results.

4.4. RF Routing Guidelines

For user's PCB, the characteristic impedance of all RF traces should be controlled to 50 Ω. The impedance of the RF traces is usually determined by the trace width (W), the materials' dielectric constant, the height from the reference ground to the signal layer (H), and the spacing between RF traces and grounds (S). Microstrip or coplanar waveguide is typically used in RF layout to control characteristic impedance. The following are reference designs of microstrip or coplanar waveguide with different PCB structures.

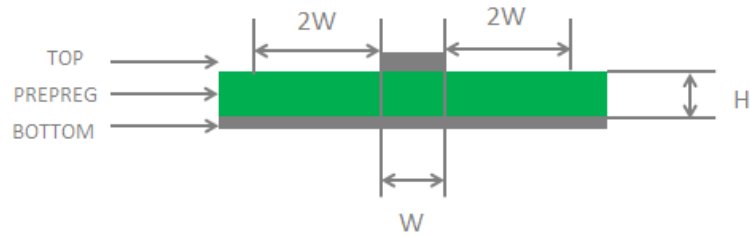


Figure 21: Microstrip on a 2-layer PCB

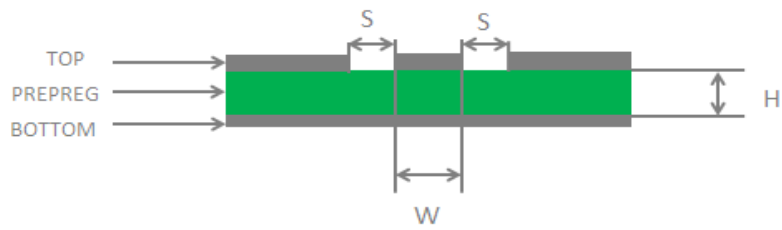


Figure 22: Coplanar Waveguide on a 2-layer PCB

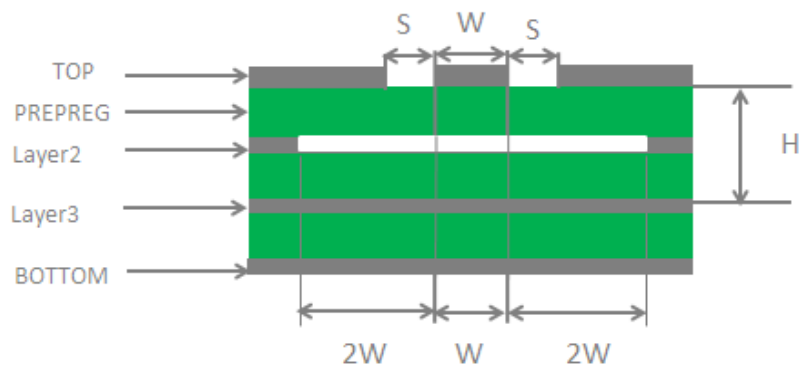


Figure 23: Coplanar Waveguide on a 4-layer PCB (Layer 3 as Reference Ground)

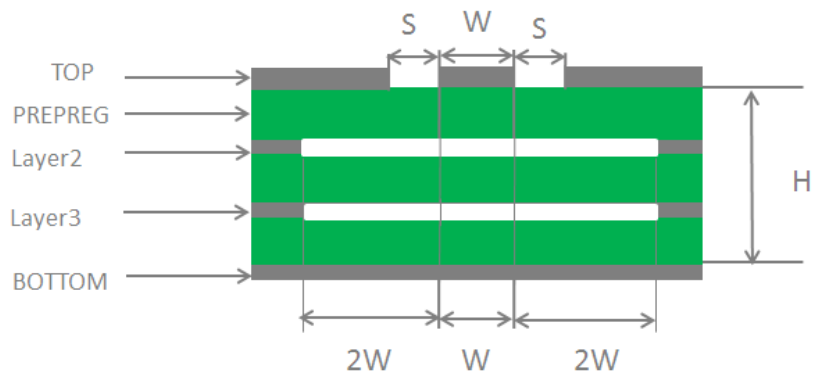


Figure 24: Coplanar Waveguide on a 4-layer PCB (Bottom Layer as Reference Ground)

To ensure reliable RF performance, the following principles should be complied with in RF layout design:

- Use an impedance simulation tool to accurately control the characteristic impedance of RF traces at 50 Ω .
- The GND pins adjacent to the RF pin should not be designed as thermal relief pads, and should be fully grounded.
- The distance between the RF pin and the RF connector should be as short as possible, and all the right-angle traces should be changed to curve ones. The recommended trace angle is 135°.
- There should be clearance under the signal pin of the antenna connector or solder joint.
- The reference ground of RF traces should be complete. Meanwhile, adding some ground vias around RF traces and the reference ground helps to improve RF performance. The distance between the ground vias and RF traces should be at least twice the width of RF signal traces ($2 \times W$).
- Keep RF traces away from interference sources (such as DC-DC, (U)SIM/USB/SDIO high frequency digital signals, display signals, and clock signals), and avoid intersection and paralleling between traces on adjacent layers.

For more details on the reference design of RF Layout, see **document [4]**.

4.5. Antenna Design Requirements

To minimize the loss on RF trace and RF cable, pay attention to the antenna design. The following tables show the requirements on an NB-IoT antenna.

Table 24: Antenna Cable Insertion Loss Requirements

Band	Requirements
LTE B5/B8/B12/B13/B17/B18/B19/B20/B28/B85	Cable insertion loss: < 1 dB
LTE B1/B2/B3/B4/B25/B66/B70	Cable insertion loss: < 1.5 dB

Table 25: Required Antenna Parameters

Parameter	Requirements
Frequency Range	698–2180 MHz
VSWR	≤ 2
Efficiency	> 30 %

Max. Input Power (W)	50
Input Impedance (Ω)	50

4.6. Transmitting Power

Table 26: Transmitting Power

Frequency Band	Max.	Min.
B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/ B19/B20/B25/B28/B66/B70/B85	23 dBm $\pm$ 2.7 dB	< -39 dBm

NOTE

The design conforms to the NB-IoT radio protocols in 3GPP Rel-13 and Rel-14.

4.7. Receiver Sensitivity

Table 27: Receiver Sensitivity without Retransmission (Throughput $\geq$ 95 %)

Frequency Band	Receiver Sensitivity	3GPP Requirement
B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/ B20/B25/B28/B66/B70/B85	-116 dBm	-107.5 dBm

Table 28: Receiver Sensitivity in 128 Retransmission (Throughput $\geq$ 95 %)

Frequency Band	Receiving Sensitivity
B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/B20/B25/B28/B66/B70/B85	$\leq$ -129 dBm

4.8. RF Connector Recommendation

If RF connector is used for antenna connection, it is recommended to use the U.FL-R-SMT connector provided by Hirose.

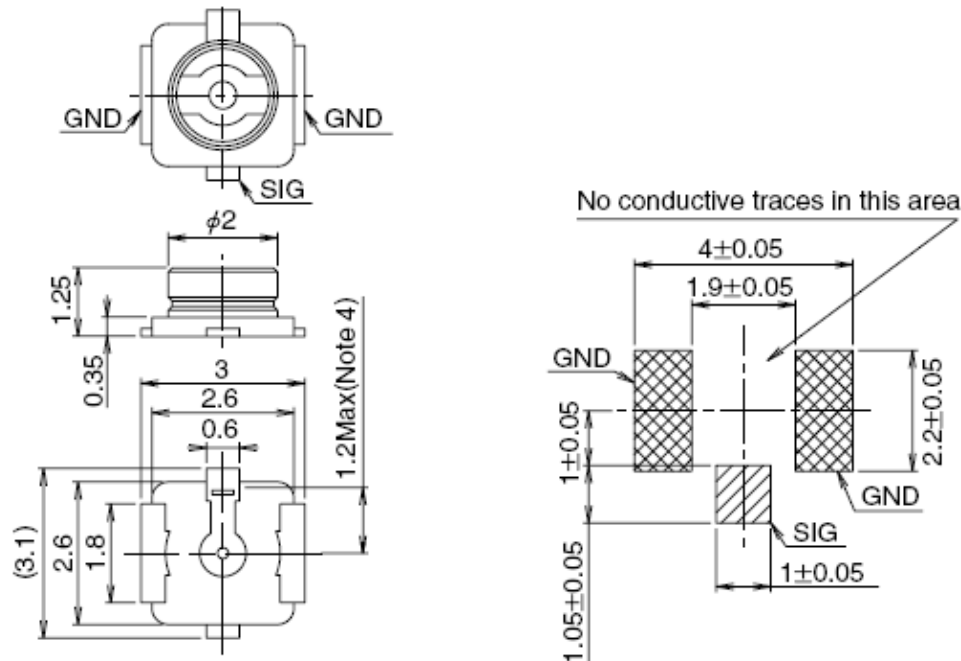


Figure 25: Dimensions of the Receptacle (Unit: mm)

U.FL-LP series mated plugs listed in the following figure can be used to match the U.FL-R-SMT.

Part No.	U.FL-LP-040	U.FL-LP-066	U.FL-LP(V)-040	U.FL-LP-062	U.FL-LP-088
Mated Height	2.5mm Max. (2.4mm Nom.)	2.5mm Max. (2.4mm Nom.)	2.0mm Max. (1.9mm Nom.)	2.4mm Max. (2.3mm Nom.)	2.4mm Max. (2.3mm Nom.)
Applicable cable	Dia. 0.81mm Coaxial cable	Dia. 1.13mm and Dia. 1.32mm Coaxial cable	Dia. 0.81mm Coaxial cable	Dia. 1mm Coaxial cable	Dia. 1.37mm Coaxial cable
Weight (mg)	53.7	59.1	34.8	45.5	71.7
RoHS	YES				

Figure 26: Specifications of Mated Plugs

The following figure describes the space factor of mated connectors.

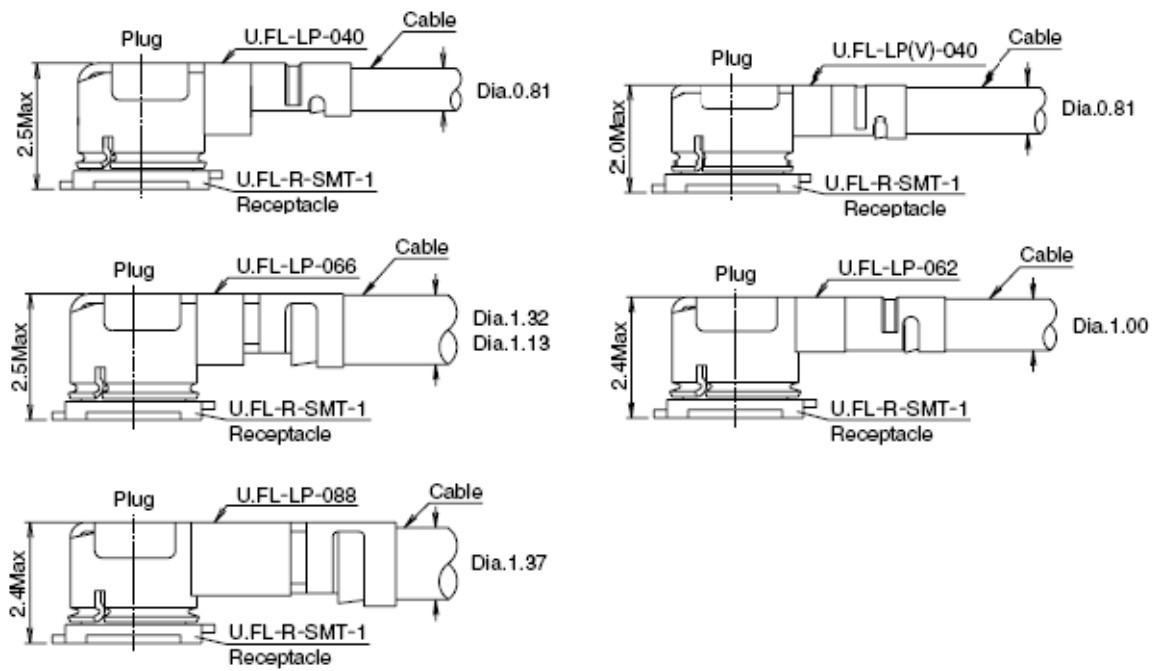


Figure 27: Space Factor of Mated Connectors (Unit: mm)

For more details, visit <http://www.hirose.com>.

5 Electrical Characteristics and Reliability

5.1. Absolute Maximum Ratings

Absolute maximum ratings for power supply and voltage on digital and analog pins of the module are listed in the following table.

Table 29: Absolute Maximum Ratings

Parameter	Min.	Max.	Unit
V _{BAT}	-0.3	4.5	V
Voltage at Digital Pins	-0.3	3.6	V
Voltage at Analog Pins	-0.3	3.6	V

5.2. Operating and Storage Temperatures

The following table lists the operating and storage temperatures of the module.

Table 30: Operating and Storage Temperatures

Parameter	Min.	Typ.	Max.	Unit
Operating Temperature Range ⁶	-35	+25	+75	°C

⁶ Within this range, the module's indicators comply with 3GPP specification requirements.

Extended Temperature Range ⁷	-40	-	+85	°C
Storage Temperature Range	-40	-	+90	°C

5.3. Power Consumption

The table below lists the power consumption of BC660K-GL under different states.

Table 31: Module Power Consumption (3.3 V VBAT Power Supply)

Deep Sleep					
AP Mode	Modem Mode	Min.	Avg.	Max.	Unit
Idle	PSM	-	0.8	-	μA
Light Sleep					
AP Mode	Modem Mode	Min.	Avg.	Max.	Unit
	eDRX = 40.96 s, PTW = 10.24 s, ECL = 0	-	38	-	μA
Idle	@ DRX = 1.28 s, ECL = 0	-	220	-	μA
	@ DRX = 2.56 s, ECL = 0	-	110	-	μA
Active ⁸					
AP Mode	Modem Mode	Min.	Avg.	Max. ⁹	Unit
	B1 @ 23.4 dBm	-	111	300	mA
	B2 @ 23.4 dBm	-	104	308	mA
	B3 @ 23.2 dBm	-	95	277	mA
Normal	Single-tone (15 kHz subcarrier spacing)		94	275	mA
	B4 @ 23.2 dBm	-			

⁷ Within this range, the module retains the ability to establish and maintain functions such as SMS and data transmission, without any unrecoverable malfunction. Radio spectrum and radio network remain uninfluenced, whereas the value of one or more parameters, such as P_{out}, may decrease and fall below the range of the 3GPP specified tolerances. When the temperature returns to the normal operating temperature range, the module's indicators will comply with 3GPP specification requirements again.

⁸ Power consumption under laboratory instrument test condition.

⁹ The "maximum value" in "Active" mode refers to the maximum pulse current during RF emission.

	B5 @ 23.1 dBm	-	98	270	mA
	B8 @ 23.3 dBm	-	105	299	mA
	B12 @ 23.2 dBm	-	120	332	mA
	B13 @ 23.2 dBm	-	100	283	mA
	B17 @ 23.2 dBm	-	115	325	mA
	B18 @ 23.2 dBm	-	94	265	mA
	B19 @ 23.2 dBm	-	95	270	mA
	B20 @ 23.2 dBm	-	98	272	mA
	B25 @ 22.8 dBm	-	108	301	mA
	B28 @ 23.3 dBm	-	109	310	mA
	B66 @ 23.2 dBm	-	95	277	mA
	B70 @ 23.6 dBm	-	98	287	mA
	B85 @ 23.2 dBm	-	115	329	mA
	B1 @ 23.4 dBm	-	268	347	mA
	B2 @ 23.5 dBm	-	256	332	mA
Single-tone (3.75 kHz subcarrier spacing)	B3 @ 23.6 dBm	-	228	295	mA
	B4 @ 23.6 dBm	-	225	291	mA
	B5 @ 23.7 dBm	-	222	290	mA
	B8 @ 23.4 dBm	-	239	311	mA
	B12 @ 23.4 dBm	-	271	349	mA
	B13 @ 23.8 dBm	-	228	293	mA
	B17 @ 23.4 dBm	-	267	344	mA
	B18 @ 23.7 dBm	-	219	285	mA
	B19 @ 23.6 dBm	-	220	286	mA
	B20 @ 23.6 dBm	-	222	290	mA

B25 @ 23.6 dBm	-	257	331	mA
B28 @ 23.5 dBm	-	256	330	mA
B66 @ 23.6 dBm	-	227	293	mA
B70 @ 23.6 dBm	-	222	287	mA
B85 @ 23.4 dBm	-	270	347	mA

NOTE

The above power consumption data are tested under 50 Ω impedance.

5.4. Digital I/O Characteristics

Table 31: USIM I/O Requirements

Parameter	Description	Min.	Max.	Unit
V_{IH}	High-level input voltage	$0.7 \times \text{USIM_VDD}$	-	V
V_{IL}	Low-level input voltage	-	$0.2 \times \text{USIM_VDD}$	V
V_{OH}	High-level output voltage	$0.8 \times \text{USIM_VDD}$	-	V
V_{OL}	Low-level output voltage	-	$0.15 \times \text{USIM_VDD}$	V

Table 32: Other I/O Requirements

Parameter	Description	Min.	Max.	Unit
V_{IH}	High-level input voltage	$0.7 \times \text{VDD_EXT}$	-	V
V_{IL}	Low-level input voltage	-	$0.2 \times \text{VDD_EXT}$	V
V_{OH}	High-level output voltage	$0.8 \times \text{VDD_EXT}$	-	V
V_{OL}	Low-level output voltage	-	$0.15 \times \text{VDD_EXT}$	V

5.5. ESD Protection

Static electricity occurs naturally and it may damage the module. Therefore, applying proper ESD countermeasures and handling methods is imperative. For example, wear anti-static gloves during the development, production, assembly and testing of the module; add ESD protection components to the ESD sensitive interfaces and points in the product design.

Table 33: Electrostatic Discharge Characteristics (Temperature: 25–30 °C, Humidity: 40 ±5 %)

Tested Interface	Contact Discharge	Air Discharge	Unit
VBAT, GND	±5	±10	kV
Antenna interface	±5	±10	kV
Other interfaces	±0.5	±1	kV

6 Mechanical Information

This chapter describes the mechanical dimensions of the module. All dimensions are measured in millimeter (mm), and the dimensional tolerances are ± 0.2 mm unless otherwise specified.

6.1. Mechanical Dimensions

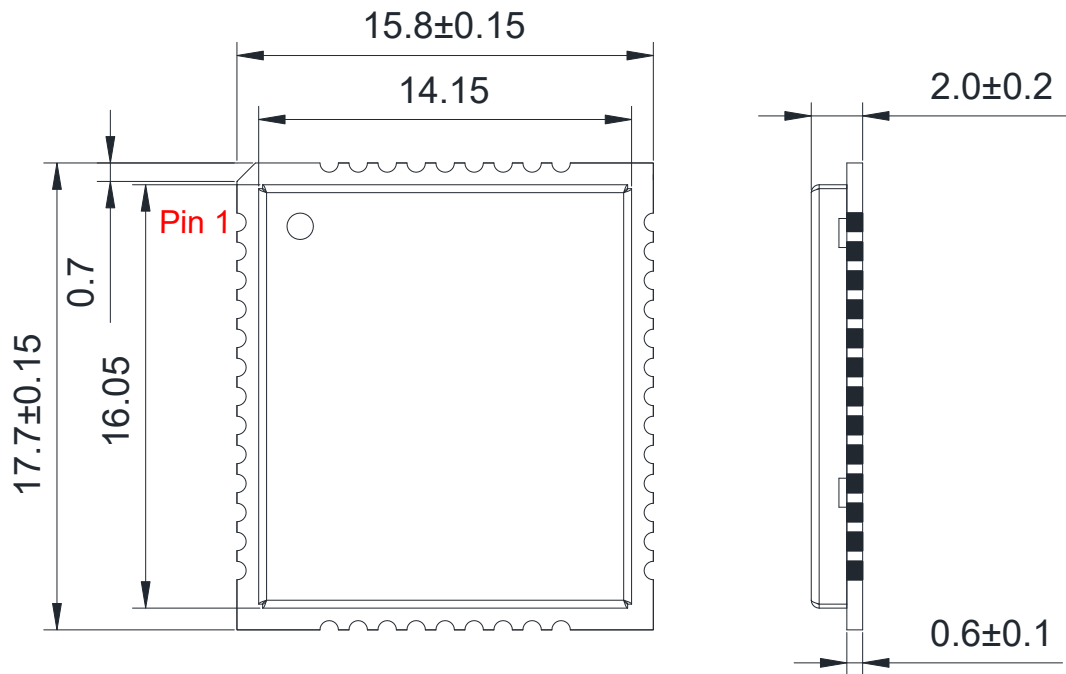


Figure 28: Module Top and Side Dimensions

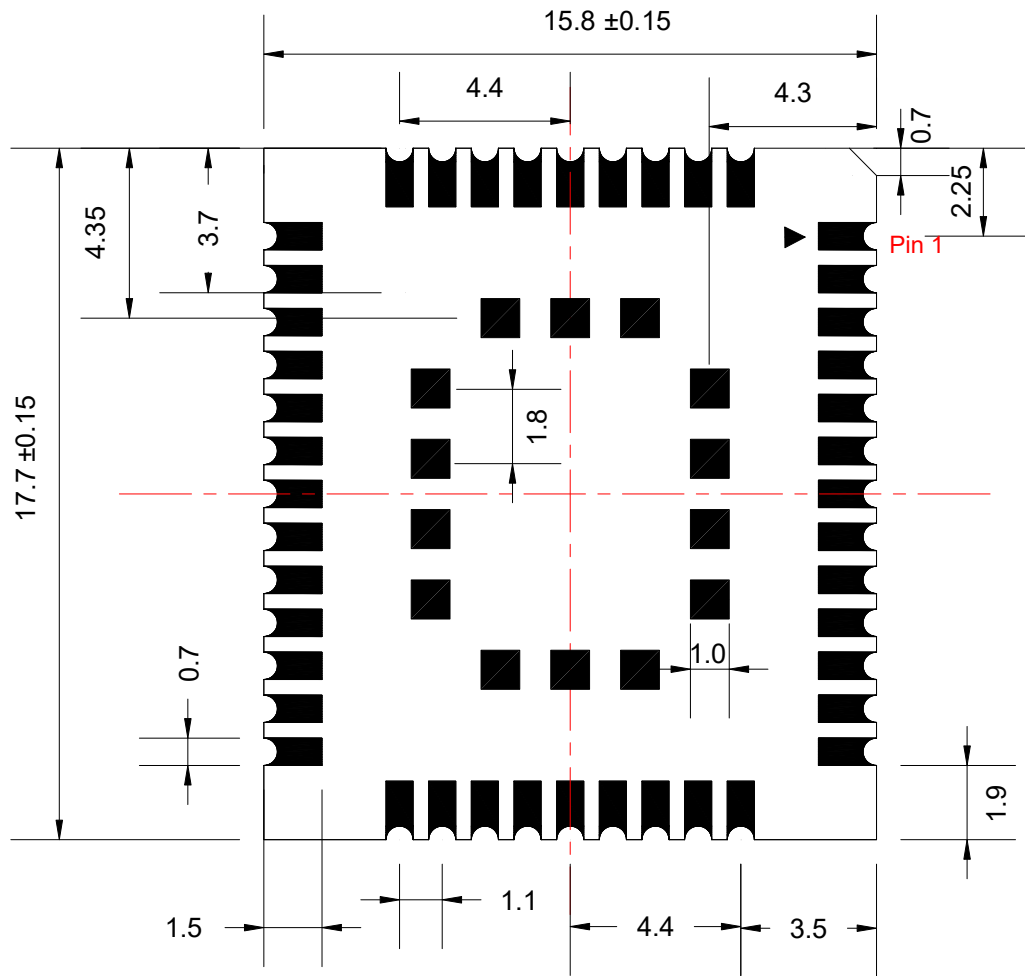


Figure 29: Module Bottom Dimensions

NOTE

The module's coplanarity standard: ≤ 0.13 mm.

6.2. Recommended Footprint

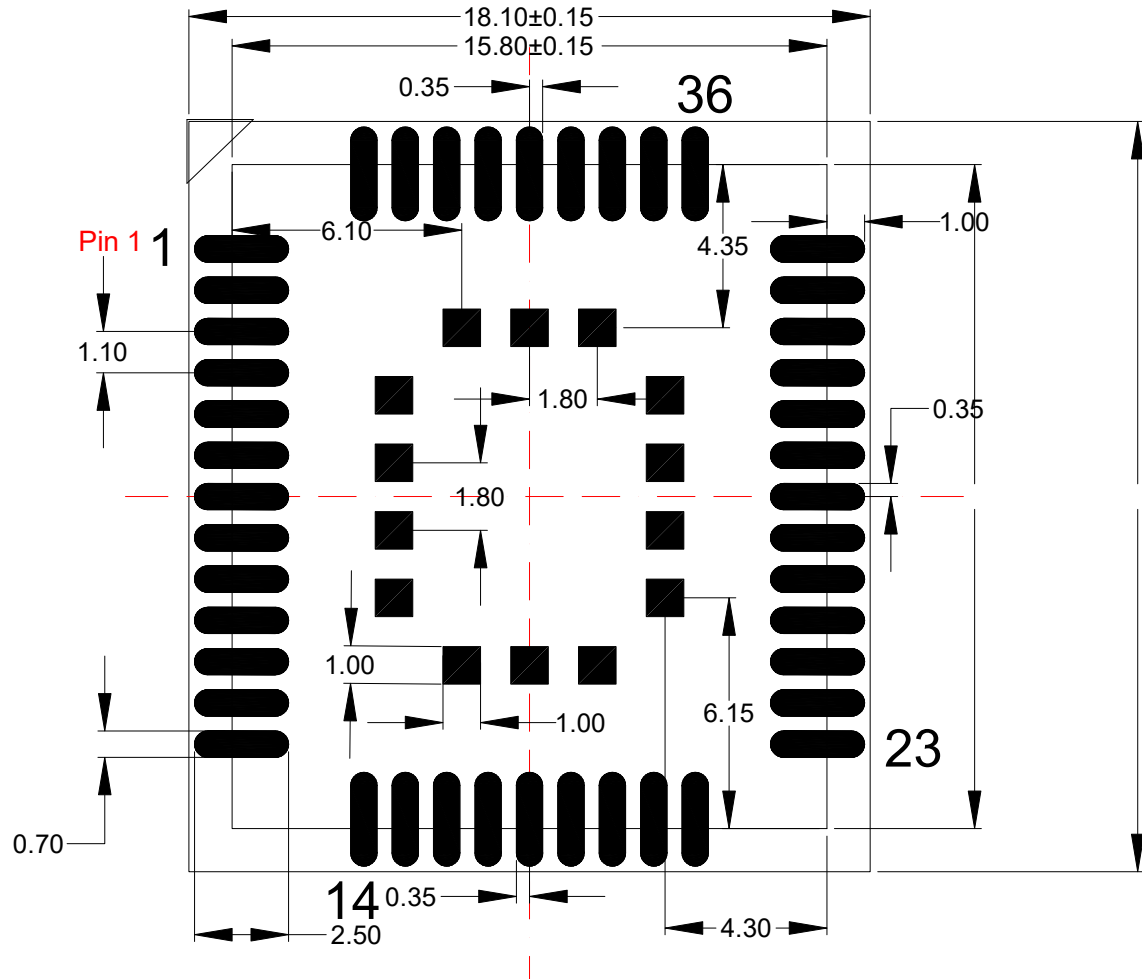


Figure 30: Recommended Footprint

NOTE

Keep at least 3 mm between the module and other components on the motherboard to improve soldering quality and maintenance convenience.

6.3. Top and Bottom Views

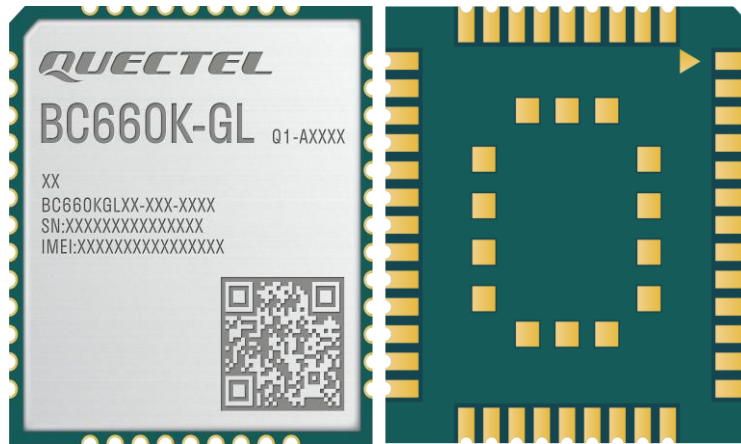


Figure 31: Top and Bottom Views of the Module

NOTE

Images above are for illustration purpose only and may differ from the actual module. For authentic appearance and label, please refer to the module received from Quectel.

7 Storage, Manufacturing and Packaging

7.1. Storage Conditions

The module is provided with vacuum-sealed packaging. MSL of the module is rated as 3. The storage requirements are shown below.

1. Recommended Storage Condition: the temperature should be 23 ± 5 °C and the relative humidity should be 35–60 %.
2. Shelf life (in a vacuum-sealed packaging): 12 months in Recommended Storage Condition.
3. Floor life: 168 hours ¹⁰ in a factory where the temperature is 23 ± 5 °C and relative humidity is below 60 %. After the vacuum-sealed packaging is removed, the module must be processed in reflow soldering or other high-temperature operations within 168 hours. Otherwise, the module should be stored in an environment where the relative humidity is less than 10 % (e.g., a dry cabinet).
4. The module should be pre-baked to avoid blistering, cracks and inner-layer separation in PCB under the following circumstances:
 - The module is not stored in Recommended Storage Condition;
 - Violation of the third requirement mentioned above;
 - Vacuum-sealed packaging is broken, or the packaging has been removed for over 24 hours;
 - Before module repairing.
5. If needed, the pre-baking should follow the requirements below:
 - The module should be baked for 24 hours at 120 ± 5 °C;
 - The module must be soldered to PCB within 24 hours after the baking, otherwise it should be put in a dry environment such as in a dry cabinet.

¹⁰ This floor life is only applicable when the environment conforms to *IPC/JEDEC J-STD-033*. It is recommended to start the solder reflow process within 24 hours after the package is removed if the temperature and moisture do not conform to, or are not sure to conform to *IPC/JEDEC J-STD-033*. Do not unpack the modules in large quantities until they are ready for soldering.

NOTE

1. To avoid blistering, layer separation and other soldering issues, extended exposure of the module to the air is forbidden.
2. Take out the module from the package and put it on high-temperature-resistant fixtures before baking. If shorter baking time is desired, see *IPC/JEDEC J-STD-033* for the baking procedure.
3. Pay attention to ESD protection, such as wearing anti-static gloves, when touching the modules.

7.2. Manufacturing and Soldering

Push the squeegee to apply the solder paste on the surface of stencil, thus making the paste fill the stencil openings and then penetrate to the PCB. Apply proper force on the squeegee to produce a clean stencil surface on a single pass. To guarantee module soldering quality, the thickness of stencil for the module is recommended to be 0.15–0.18 mm. For more details, see **document [5]**.

The recommended peak reflow temperature should be 235–246 °C, with 246 °C as the absolute maximum reflow temperature. To avoid damage to the module caused by repeated heating, it is recommended that the module should be mounted only after reflow soldering for the other side of PCB has been completed. The recommended reflow soldering thermal profile (lead-free reflow soldering) and related parameters are shown below.

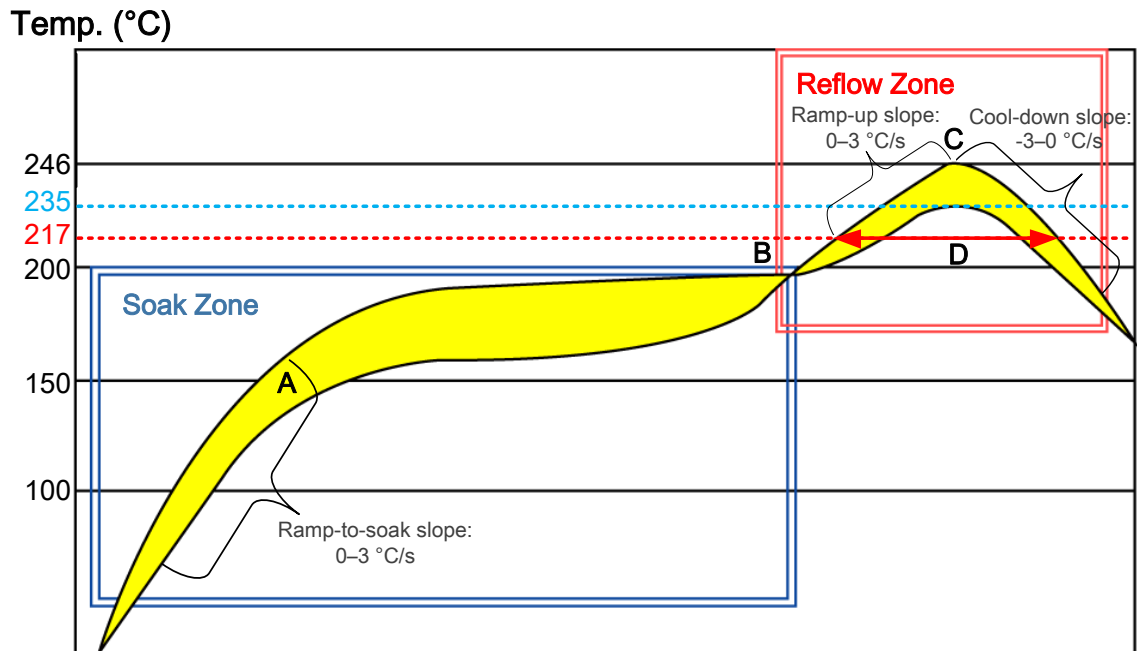


Figure 32: Recommended Reflow Soldering Thermal Profile

Table 34: Recommended Thermal Profile Parameters

Factor	Recommended Value
Soak Zone	
Ramp-to-soak slope	0–3 °C/s
Soak time (between A and B: 150 °C and 200 °C)	70–120 s
Reflow Zone	
Ramp-up slope	0–3 °C/s
Reflow time (D: over 217°C)	40–70 s
Max. temperature	235–246 °C
Cool-down slope	-3–0 °C/s
Reflow Cycle	
Max. reflow cycle	1

NOTE

1. The above profile parameter requirements are for the measured temperature of the solder joints. Both the hottest and coldest spots of solder joints on the PCB should meet the above requirements.
2. During manufacturing and soldering, or any other processes that may contact the module directly, NEVER wipe the module's shielding can with organic solvents, such as acetone, ethyl alcohol, isopropyl alcohol, trichloroethylene, etc. Otherwise, the shielding can may become rusted.
3. The shielding can for the module is made of Cupro-Nickel base material. It is tested that after 12 hours' Neutral Salt Spray test, the laser engraved label information on the shielding can is still clearly identifiable and the QR code is still readable, although white rust may be found.
4. If a conformal coating is necessary for the module, do NOT use any coating material that may chemically react with the PCB or shielding cover, and prevent the coating material from flowing into the module.
5. Avoid using ultrasonic technology for module cleaning since it can damage crystals inside the module.
6. Avoid using materials that contain mercury (Hg), such as adhesives, for module processing, even if the materials are RoHS compliant and their mercury content is below 1000 ppm (0.1 %).
7. Corrosive gases may corrode the electronic components inside the module, affecting their reliability and performance, and potentially leading to a shortened service life that fails to meet the designed lifespan. Therefore, do not store or use unprotected modules in environments containing corrosive gases such as hydrogen sulfide, sulfur dioxide, chlorine, and ammonia.
8. Due to the complexity of the SMT process, please contact Quectel Technical Support in advance for

any situation that you are not sure about, or any process (e.g. selective soldering, ultrasonic soldering) that is not mentioned in **document [6]**.

7.3. Packaging Specification

This chapter outlines the key packaging parameters and processes. All figures below are for reference purposes only, as the actual appearance and structure of packaging materials may vary in delivery.

The modules are packed in a tape and reel packaging as specified in the sub-chapters below.

7.3.1. Carrier Tape

Carrier tape dimensions are illustrated in the following figure and table:

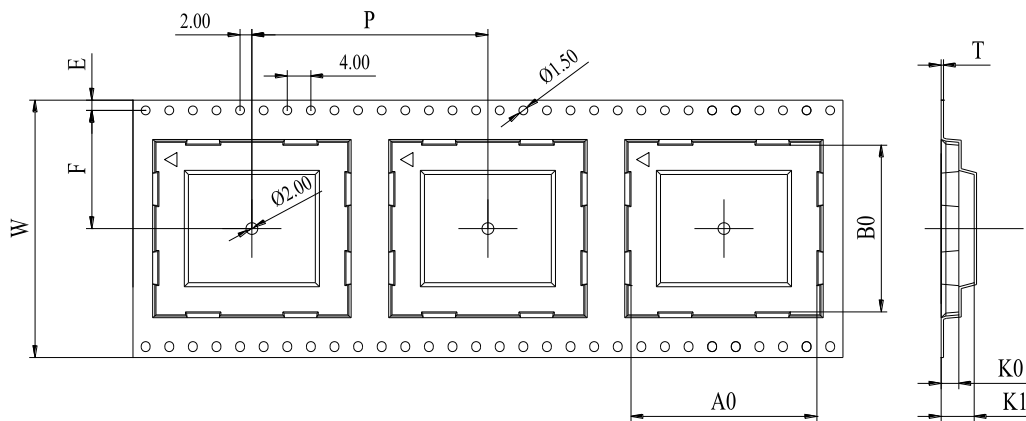


Figure 33: Carrier Tape Dimension Drawing (Unit: mm)

Table 35: Carrier Tape Dimension Table (Unit: mm)

W	P	T	A0	B0	K0	K1	F	E
32	24	0.35	16.2	18.1	2.9	3.4	14.2	1.75

7.3.2. Plastic Reel

Plastic reel dimensions are illustrated in the following figure and table:

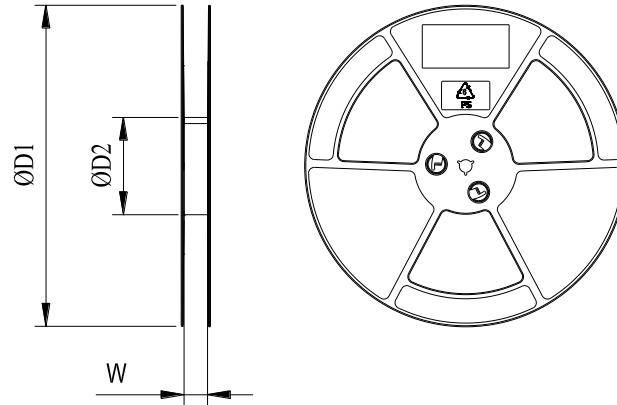


Figure 34: Plastic Reel Dimension Drawing

Table 36: Plastic Reel Dimension Table (Unit: mm)

ØD1	ØD2	W
380	100	32.5

7.3.3. Mounting Direction

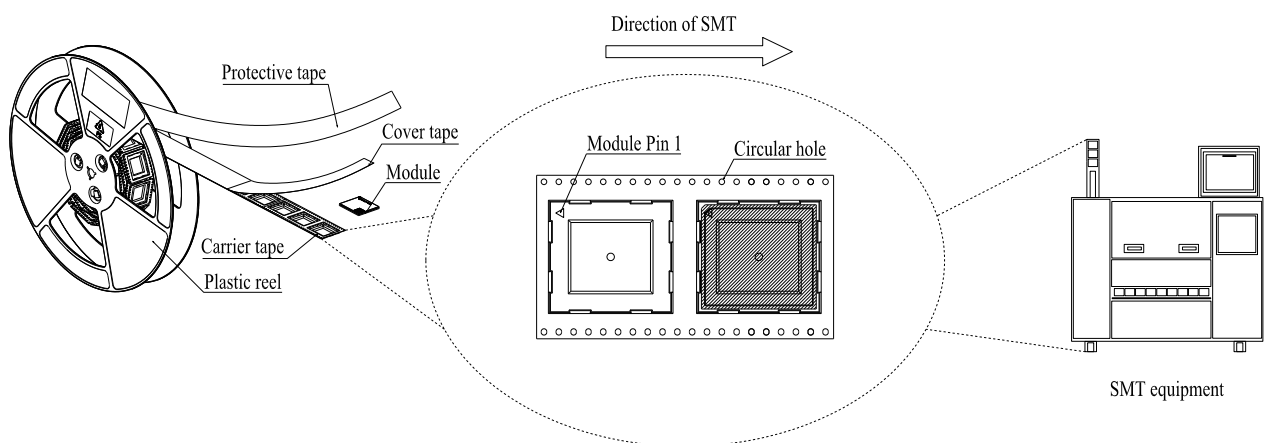
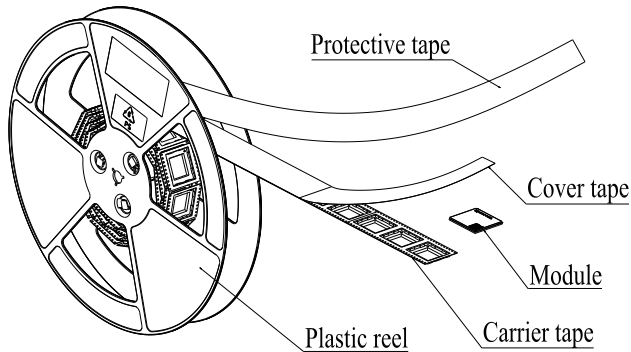


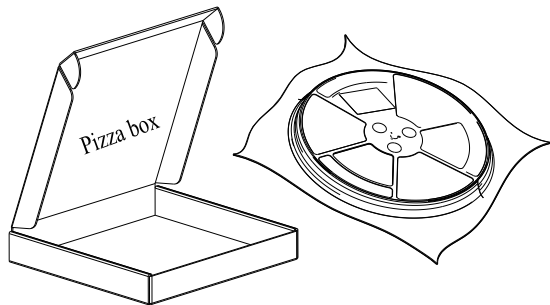
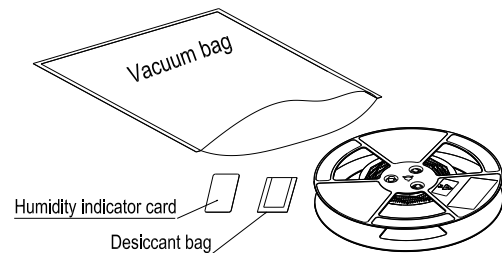
Figure 35: Mounting Direction

7.3.4. Packaging Process



Place the modules onto the carrier tape cavity and cover them securely with cover tape. Wind the heat-sealed carrier tape onto a plastic reel and apply a protective tape for additional protection. 1 plastic reel can pack 1000 modules.

Place the packaged plastic reel, humidity indicator card and desiccant bag into a vacuum bag, and vacuumize it.



Place the vacuum-packed plastic reel into a pizza box.

Place the 4 packaged pizza boxes into 1 carton and seal it. 1 carton can pack 4000 modules.

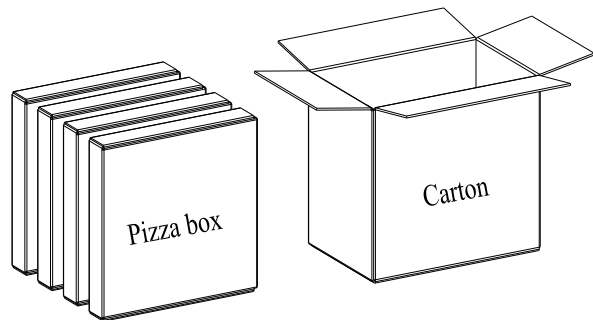


Figure 36: Packaging Process

8 Appendix References

Table 37: Related Documents

Document Name
[1] Quectel_BC660K-GL-TE-B_User_Guide
[2] Quectel_BC660K-GL&BC950K-GL_AT_Commands_Manual
[3] Quectel_BC660K-GL_Reference_Design
[4] Quectel_RF_Layout_Application_Note
[5] Quectel_Module_Stencil_Design_Requirements
[6] Quectel_Module_SMT_Application_Note

Table 38: Terms and Abbreviations

Abbreviation	Description
ADC	Analog-to-Digital Converter
AP	Application Processor
CPU	Central Processing Unit
DRX	Discontinuous Reception
DTLS	Datagram Transport Layer Security
ECL	Enhanced Coverage Level
eDRX	extended Discontinuous Reception
EMI	Electromagnetic Interference
ESD	Electrostatic Discharge

ESR	Equivalent Series Resistance
HD-FDD	Half Frequency Division Duplexing
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol over Secure Socket Layer
I/O	Input/Output
I _{PP}	Peak Pulse Current
I _{BR}	Breakdown Current
kbps	Kilobits per second
LCC	Leadless Chip Carrier (package)
LDO	Low-dropout Regulator
LED	Light Emitting Diode
LGA	Land Grid Array
LTE	Long Term Evolution
LwM2M	Lightweight M2M
MSL	Moisture Sensitivity Levels
MQTT	Message Queuing Telemetry Transport
NB-IoT	Narrow Band-Internet of Things
OC	Open Collector
OD	Open Drain
PC	Personal Computer
PCB	Printed Circuit Board
PDN	Public Data Network
PMU	Power Management Unit
PSM	Power Save Mode
PTW	Paging Time Window

RF	Radio Frequency
RTC	Real Time Clock
RXD	Receive Data
SMD	Surface Mount Device
SMS	Short Message Service
TAU	Tracking Area Update
TCP	Transmission Control Protocol
TE	Terminal Equipment
TLS	Transport Layer Security
TXD	Transmitting Data
UART	Universal Asynchronous Receiver & Transmitter
UDP	User Datagram Protocol
UE	User Equipment
URC	Unsolicited Result Code
(U)SIM	Universal Subscriber Identification Module
VSWR	Voltage Standing Wave Ratio
V _{ESD}	Electrostatic Discharge Voltage
V _{max}	Maximum Voltage
V _{nom}	Nominal Voltage
V _{min}	Minimum Voltage
V _{IHmax}	Maximum High-level Input Voltage
V _{IHmin}	Minimum High-level Input Voltage
V _{ILmax}	Maximum Low-level Input Voltage
V _{ILmin}	Minimum Low-level Input Voltage
V _I max	Absolute Maximum Input Voltage

$V_{I\text{nom}}$	Absolute Nominal Input Voltage
$V_{I\text{min}}$	Absolute Minimum Input Voltage
$V_{OH\text{max}}$	Maximum High-level Output Voltage
$V_{OH\text{min}}$	Minimum High-level Output Voltage
$V_{OL\text{max}}$	Maximum Low-level Output Voltage
$V_{OL\text{min}}$	Minimum Low-level Output Voltage
V_{RWM}	Working Peak Reverse Voltage